

*EXAMINING CHANGES IN MOUSE CORTICAL NETWORK
DYNAMICS DUE TO SLEEP DEPRIVATION*

A Thesis

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ABSTRACT

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Sleep is an essential biological function, and those who get too little invariably suffer cognitive impairments that frequently lead to accidents. While the responses of brain-wide networks and handfuls of cells to sleep deprivation have been studied extensively, studies of local clusters of tens to hundreds of cells have been relatively rare. Here we use *in vivo* fluorescent calcium imaging to examine the response of the medial prefrontal cortex to six hours of sleep restriction. We examine the changing activity levels of individual cells, the evolution of pairwise correlations, and the effect of distance on said correlations. We specifically examine the changing dynamics during the minute and a half leading up to a sleep state transition by breaking the fluorescence traces into ten-second chunks and plotting the outcome measures of each successive chunk, a method known as trajectory analysis. This analysis resulted in both expected and unexpected results. Event rate and pairwise correlation increased after sleep deprivation in line with hypotheses. Neither event rate nor pairwise correlation increased in the moments leading up to sleep onset unlike we hypothesized. The pairwise correlation differences were unexpectedly larger in distant cell pairs than close cell pairs. This study also offers thorough examination of fluorescent imaging methods and methodological suggestions for similar future studies.

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INTRODUCTION

Sleep is an essential and ubiquitous biological function. It can most simply be defined as a period of behavioral quiescence that recurs about once a day. Under this definition (or, when available, a brain electrical activity definition), every single animal ever observed sleeps (with at least a part of its brain at a time) in some form or another (1). Adequate sleep promotes proper bodily functions, but, unfortunately, Americans often get less than adequate sleep. According to a 2022 survey, every single state reported at least 1 in 4 adults getting less than seven hours of sleep per night (2). This causes significant attention, reaction time, and cognitive function deficits in both humans and animals (3-6). The exploration of disordered sleep benefits from an understanding of natural sleep. Sleep can be measured using similar methods in humans and animals, opening an avenue of study.

Electroencephalography (EEG) is a test in which electrodes are placed around the skull to measure electrical changes due to the neurons below. It is used to identify sleep stages across organisms. It is usually combined with an electromyogram (EMG) in rodents. In humans, it is combined with EMG, plus heart rate, breathing rate, eye movement, and other physiological measures to compile a polysomnogram, a common diagnostic tool. While the length of a sleep cycle differs significantly between rodents and humans, the EEG trace qualities emblematic of each vigilance state within the cycle remain the same. The three basic vigilance states are wake, rapid eye movement (REM), and non-REM (NREM) sleep. REM is the shortest and least common state, and we do not have enough recorded to talk about it at length in this report. Wake EEG traces contain predominantly low-amplitude, high-frequency waves (13-30Hz beta waves), while NREM sleep is marked by the emergence of high-amplitude, low-frequency waves (1-4Hz delta waves) that only get higher and slower as the state deepens (7). High amplitude EEG waves are associated with local neuronal synchrony, and low amplitude waves are associated with desynchrony (8,9), since a single electrode measures the combined activity of millions of neurons. Some is known about the activity of individual neurons in the various sleep conditions as well. Neurons express a

higher variation in firing rate during NREM compared to the other states (10), and sleep need increases the neuron firing rate in NREM compared to the same state when sleep replete (11,12). Studies involving local field potential (LFP) and multiunit activity (MUA) have been used to extensively examine local networks on the order of several to about ten neurons, while fMRI has been used to examine network changes across the whole brain, revealing aberrant connectivity between brain regions (13). Relatively few, however, have used fluorescent microendoscopy to examine local networks on the order of several tens to a few hundred neurons. This disparity should be remedied, especially considering the burgeoning concept of local sleep.

MUA recordings can be used to isolate individual cell activity within a local cluster of neurons. When EEG detects high amounts of slow wave activity, the cell synchrony that creates those slow waves can be seen directly using MUA. The cortex can be divided into “on” periods in which the population of neurons around the microelectrode arrays is active and “off” periods in which none of the neurons in range fire at all. The frequency of “off” periods is generally higher in NREM than in wake, but increases with sustained wakefulness and decreases throughout sleep, mirroring the patterns of delta waves, leading researchers to conclude that the two phenomena are intrinsically linked, with increasing delta waves and “off” periods during wake seen as intrusions of a sleep-like state and their decrease during NREM seen as alleviation of sleep pressure (11,14,15). More recent observations in living animals have revealed that “off” periods can vary greatly by brain region and are promoted by preceding activity of the observed region (16,17). This regional variability is a key feature of local sleep, in which cortical areas can transition independently into sleep-like states (18,19). It is still unclear, however, whether functional heterogeneity exists within brain regions or only between them, and whether all brain regions are susceptible to the phenomenon. Sleep deprivation could disrupt local neuronal networks, but traditional methods of cortical activity measurement lack the field of view to determine whether or not it does. Fortunately, new tools enable us to better examine this. Fluorescent microendoscopy enables simultaneous recording of the activity and location of tens to hundreds of neurons in freely behaving animals, bridging the gap between single-cell electrophysiology and whole brain imaging.

“On” and “off” are bistable states in cortical neurons. The mechanism underlying “off” period initiation involves both permissive and active components. The classical account attributes “off” period initiation with a concept known as “disfacilitation,” in which reduced acetylcholine and norepinephrine signaling lead to increased potassium leak and hyperpolarization increasing the tendency for cells to come to rest in the “off” state (20). Newer research identifies inhibition of pyramidal cells by Somatostatin-expressing Martinotti cells as a potential driving force for the

hyperpolarization and the resultant transition to “off” periods (21). Once initiated, the “off” period is sustained by synaptic depression and potassium adaptation currents until rebound sodium and hyperpolarization-activated cation currents allow the neuron to return to the “on” state (20). Very recent work has demonstrated that artificially inducing “on” and “off” periods during wakefulness is sufficient to reduce local sleep pressure, trigger synaptic downscaling and promote memory consolidation, establishing that these dynamics underlie at least some core functions of sleep independent of global sleep state (22).

This study targets the mouse medial prefrontal cortex (mPFC), specifically the prelimbic (PL) region. Although there are some irregularities in rodent PFC naming (as discussed in (23)), the region can be considered homologous to the human dorsal anterior cingulate cortex (dACC) region (24). Both regions are within their respective species’ prefrontal cortex (PFC), which broadly handles working memory, abstract reasoning, decision-making, planning, judgment, and conscious control of attention, emotion, and action (25). The human dACC (also known as the pregenual anterior cingulate area 32 or A32) is involved in decision-making, attention, and error monitoring specifically (26), while the rodent PL handles attention, emotional regulation, and reward-based decision making particularly (27). Both humans and rodents experience significant attentional dysfunction after sleep restriction (28), which could be related to metabolic changes found in the human PFC (29,30). The mouse mPFC is also an easy target for microendoscopy because it is so superficial anatomically, sitting about 2 millimeters below the skull. These factors make the rodent prelimbic mPFC a good target for examining the human cognitive dysfunctions that lead to accidents after sleep loss.

The mPFC is also well-positioned within the cortical sleep network. Large-scale EEG recordings demonstrate that slow oscillations propagate predominantly in an anterior-to-posterior direction across the cortex (31), a pattern corroborated at finer spatial scales by electrocorticographic (ECoG) recording evidence of elevated convergence and divergence activity in prefrontal cortical regions (32). Computational modeling suggests that this directional preference is not intrinsic to prefrontal cortex but instead emerges from heterogeneities in connection strengths within the human connectome, which determine which regions transition to a state dominated by slow waves first (33). The cellular-scale spatial resolution described here could provide complementary data to previous network-level studies of sleep onset and local sleep dynamics.

The overall question we’re asking is **how individual cell dynamics evolve into group dynamics**. We will be looking specifically at how quickly sleep onset occurs, how tightly

correlated local networks are, and where cell synchronization starts. We will do this with a series of analyses measuring event rates for each ROI (region of interest) and pairwise correlations in each unique pair of ROIs in the field of view, both constrained to the period leading up to transitions. After that, we will separate the pairwise correlations into distance bins to see whether there are key distances over which the ROIs synchronize. Finally, we will examine how these network dynamics respond to disruption via sleep deprivation by repeating the measurements during the same circadian phase but after six hours of vigilance maintenance by gentle handling.

This study is an attempt to generalize established local sleep findings in a new area (mPFC) and on a new scale (tens of neurons). We hypothesize that mPFC network patterns during sleep will reflect other regions' patterns, especially at the pivotal sleep/wake transitions. We expect the event rate of our putative pyramidal cells to decrease before wake to NREM transitions (34) and for sleep deprivation to result in increased event rate during NREM (11). We also expect the pairwise correlation of our cells to increase during wake to NREM transitions and for sleep deprivation to enhance the effect (11). Finally, we expect close cells to correlate more due to local network connectivity. Our specific aims are as follows:

1. Characterize single-cell activity patterns change in the moments preceding sleep onset
2. Examine pairwise correlation structure and distance-dependence

METHODS

All animal procedures were performed in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals and were approved by the Morehouse School of Medicine Institutional Animal Care and Use Committee.

Animals

One of the mice used in this study was a male that was the offspring of two heterozygous Ai162(TIT2L-GC6s-ICL-tTA2)-D mice (Jax Strain #031562, also known as Ai162) and was determined to be heterozygous for $Igs7^{tm162.1(tetO-GCaMP6s,CAG-tTA2)Hze}$ via genotyping. The other three mice were the offspring of a TIGRE2-jGCaMP8m-IRES-tTA2-WPRE dam (Jax Strain #037718) and a B6.Cg-Tg(Camk2a-cre)T29-1Stl/J sire (Jax Strain #005359, also known as T29-1). There were two female mice and one male mouse of this cross used in this study. All transgenic strains are on a C57/BL6J background. The three dual transgenic mice inherited a gene causing expression of jGCaMP8m (a fluorescent Ca^{2+} reporter, the 8th generation medium-speed variety of GCaMP developed by HHMI's Janelia Research Campus) in cells that also express Cre recombinase (Cre), and a gene causing Cre expression alongside the alpha variant gene for Ca^{2+} /calmodulin-dependent protein kinase II (CaMKII). This caused CaMKII-positive cells, which are presumed to be pyramidal cells in the neocortex (35), to express jGCaMP8m, allowing us to use microendoscopy to visualize calcium activity in cortical neurons. The Ai162 mouse also expressed GCaMP based on CaMKII expression, but that gene modification was mediated by a viral vector (described in detail in the surgery section) instead of being congenital. All mice were housed in LD 12:12 conditions, with lights on at 6AM and off at 6PM Eastern Time (shifted to 7AM/7PM during daylight saving time). The mice were allowed food and water *ad libitum*. The dual transgenic crosses were Male 2, Female 1, and Female 2 and the Ai162 was Male 1.

Surgery

All mice were at least 10 weeks old at the time of their first surgery. Surgical techniques vary by gene insertion method but will be described in terms of common techniques with qualifiers for methods that vary. Equipment and reagent details are on the materials table (Appendix 1).

The animals were first anesthetized using isoflurane gas and prepared for surgery. They were placed in a stereotaxic alignment system (stereotax), had their scalp cleared with depilatory cream, had artificial tears applied to their eyes and whiskers, and were administered a prophylactic analgesic and anti-inflammatory solution of extended-release buprenorphine HCl, dexamethasone, and lactated ringer's solution subcutaneously. The scalp was then cut away, and the fascia were retracted with cotton-tipped applicators. A centering scope and stereotaxic alignment indicator were used to level the skull and place bregma at the rotational and translational zero point on the stereotax. Three screws with pre-soldered wire leads were implanted into the skull to be used as both EEG electrodes and miniscope stabilizers. They were implanted at the caudolateral corner of the left frontal plate, the caudolateral corner of the left parietal plate, and the caudolateral corner of the right frontal plate. This was done by first using a blunt fill needle to make a pilot hole, then driving the screws with a 2.4mm slotted screwdriver and a pair of Dumont #7 profile forceps. A stereotaxic drill fitted with a trephine bit was then used to make a 1.8 mm diameter craniotomy centered 2.1mm anterior and 0.35mm to the right of bregma (Inscopix)(36). The virally transduced mouse then received a transcranial injection of viral solution. This was achieved by inserting a neuro syringe into the center of the craniotomy, progressing slowly to the intended depth of 1.8mm below the cortical surface, infusing 240 nL of solution at a speed of 1 nL/s, and allowing the syringe to dwell 5 minutes before retracting it. This targeted precisely the region of interest, the prelimbic PFC (Inscopix)(36). We used stock concentration of ready-to-image viral solution from Inscopix. The product number can be found in Appendix 1 and Inscopix lists titer details on their product page. Next, the wire leads were soldered (using lead-free flux core solder) to the leads of sockets in a Pinnacle head mount in a custom configuration. EMG leads from the same head mount were implanted bilaterally into the neck muscles of the mice. Integrated gradient index (GRIN) lenses with a diameter of 1mm were then lowered into the center of the craniotomy and advanced to a target depth of 1.6mm below the cortical surface at a rate of 1µm/s. Representative histology is presented in Appendix 2. This position was chosen to allow an appropriate working distance of

200 μm between the end of the lens and the target cells. The hardware and surgery site were then encased in dental acrylic, leaving the hardware attachment points exposed. The mice were allowed to recover in a clean cage and monitored until they reached a fully ambulatory state. The mice received supportive dexamethasone and Lactated Ringer's solutions for at least three days post operation, with the extended-release buprenorphine re-administered and subcutaneous fluid support continued until the mice returned to thriving condition as per IACUC guidelines.

There were pilot animals that used similar surgical methods but targeted the anterior cingulate area (ACA). While these animals can inform methods recommendations, they are excluded from the main analyses due to the functional heterogeneity of the regions.

Reporter

The viral vector and transgenic construct encode different versions of the reporter protein GCaMP. The three dual transgenic cross mice expressed jGCaMP8m, while the virally transduced mouse expressed GCaMP6f. The speed of the two were compared in an experiment at HHMI's Janelia Research Campus invoking 10 successive action potentials in cultured neurons (37). The onset of jGCaMP8m was significantly faster than that of GCaMP6f. The offset speed of the two was comparable. The same group invoked 160 successive action potentials in the same neurons to measure peak fluorescence over baseline in the same experiment (37). The GCaMP6f proteins gained significantly more fluorescence in that condition. See the aforementioned source for more details, as they present clear and user-friendly interactive graphs on the topics.

Experiment

All mice were given at least two weeks between surgery and experimentation. Mice were transferred to a sleep chamber and allowed to habituate for three days before the experiment. The sleep chamber environment consisted of an open-top acrylic cage with litter, bedding material, and food and water *ad libitum*. The sleep chamber light cycle was LD 12:12 (12 hours each of light and darkness), with lights on at 6AM and off at 6PM Eastern Time (shifted to 7AM/7PM during daylight saving time) to align with the housing conditions. The mice were tethered to both the miniscope and the EEG data acquisition device (DAQ) for the entirety of their time in the sleep chamber. The miniscope was tethered to a disabled Inscopix commutator, which was then

connected to a DAQ that was specific to the miniscope. The EEG head mount was tethered to a preamp attached to a Pinnacle commutator attached to the Pinnacle DAQ. The mouse was given just enough slack in the tethers to be able to reach all corners of the cage, and no tangling occurred. The Pinnacle and Inscopix DAQs were also connected to each other via a BNC cable attached to the sync port on the Inscopix DAQ and the GPIO2 port in the Pinnacle DAQ.

The experiment was conducted over three days. The first day was a baseline recording, in which one hour of simultaneous 10Hz microendoscopic fluorescent video and EEG were collected from ZT6 to ZT7. The next morning, the mouse was sleep restricted via gentle handling from ZT0 (Zeitgeber time 0, or the start of the first active hour) to ZT6 (the sixth active hour) and another hour of simultaneous video and EEG was collected ZT6-ZT7. The mouse was left undisturbed to get recovery sleep until ZT6 the next day, at which point the final hour of recording was collected. TTL pulses were used to synchronize the video and EEG recording with one tenth of a second precision. These days are referred to as Baseline, Sleep Deprivation, and Recovery respectively. Mice were anesthetized with isoflurane and euthanized for histology after the completion of the experiment. Representative histology is shown in Appendix 2. We restricted the scope of this project to the first two days after recording, but the recovery day data were plotted in Appendix 7 and they still exist for future analysis.



Figure 1: Experiment Schedule

A chart representing the timeline for any given experimental run. Each consecutive day is represented by a row, and each hour is represented by a column, with numbers in each box indicating the end of that column's period. The boxes above the Zeitgeber time row show which of the hours are during the light phase and which are during the dark phase. Colors indicate which experimental phase occurs at each hour of each day, with periods that the mouse is undisturbed during its rest period in blue, periods that it is kept awake during its rest period in red, and the mouse's active period in yellow. Green indicates simultaneous EEG and miniscope recordings, which happen during an hour of the rest phase in which the animal is undisturbed.

Data Processing

The scripts used for this project are available via github: <https://github.com/DelaneyMBeckner/DMB-Thesis-MSM-2025>. Data CSVs and TIFFs are available upon request.

Preprocessing

EEG traces were first aligned by trimming the beginning of the trace to the TTL pulse representing the first frame of miniscope recording and the end of the trace to one hour later. The traces were then broken into 360 ten-second epochs each. **These epochs are the base time unit of subsequent analyses.** The epochs were then scored manually in Pinnacle's Sirenia Sleep software by Luis Sanchez of Dr. Chris Ehlen's lab. Finally, the epoch scores (i.e. vigilance state codes) and their time stamps were exported for downstream analysis.

Miniscope recordings were preprocessed in Inscopix Data Processing Software (IDPS). They were first trimmed to be exactly 1 hour long. Then, a spatial bandpass filter that arithmetically combines each frame with versions of that frame with Gaussian blur applied was used to eliminate spatial fluctuations of less than 0.005 per pixel or more than 0.500 per pixel. Motion correction was then conducted, referencing the mean projection of the videos for frame translation and utilizing empty border pixels where necessary to preserve dimensions. Finally, an algorithm that compares each frame to the mean projection and returns the fold change was used to convert the videos to $\Delta F/F$. A researcher then scanned the video for cell-like dynamics and drew ROI (region of interest) contours manually. ROI maps for all animals can be found in Appendix 3. The same contour defining the ROI map was used for each recording of the same animal, allowing cross-day comparison of the same cells. The software then checked each ROI's fluorescence trace for any single frame increases of more than four times the median absolute deviation of the whole trace and marked them as events. A TIFF of the ROI map contour; a CSV of the event times, their corresponding ROI, and the fluorescence peak; and a CSV containing the fluorescence value for every ROI at every frame (i.e. the fluorescence traces) were exported for further analysis.

Main Analysis

All analyses were conducted in R version 4.5.0 and the Anaconda3 distribution of Python. Data cleaning was done before all analyses to remove any ROI with less than 1 event for every 18 epochs (about the length of a mouse sleep cycle) during the baseline recording. No more than 15%

of ROIs were removed for any given animal, and a full filtering report can be found in Appendix 4.

We ran signal-to-noise ratio analysis on the fluorescence traces to confirm recording quality. We defined signal as the average $\Delta F/F$ of highest values of each trace within 2 seconds of an automatically detected event. We defined noise as the standard deviation of all portions of each trace greater than two seconds away from any automatically detected event. These components and their quotients were calculated and plotted for each animal.

Our synchronized measures enable analysis on specific time frames relative to sleep stage transitions. Since wake to NREM transitions (sleep onsets) represent a point of critical sleep pressure where network changes lead to behavioral changes, we examined a window leading up to sleep onset. NREM to wake changes also represent a critical point of sleep pressure decrease that causes the pattern to reverse. Wake to NREM changes are the focus of this study, but the NREM to wake transition data plots are included in Appendix 7 for those curious.

This study uses a new cell behavior analysis paradigm known as trajectory analysis to examine the changes in ROI activity leading up to transitions between wake and NREM sleep. Transitions to and from REM were excluded. The first iteration of this analysis (and the one used for explanation) was event rate analysis, since it is a simple way to examine the trends in activity leading up to transition for each cell. The $\Delta F/F$ fluorescence traces were first separated into epochs tagged with their sleep states, then all transitions were identified, and windows with the requisite number of preceding epochs in the pre-transition state (9 for the analyses used in this study or 3 for the example in Figure 2) were extracted to get an analysis window. The number and type of transitions extracted from each recording is listed in Appendix 5. Next, the number of events detected in all traces in each epoch leading up to the transition was averaged and plotted. In the example in Figure 2, the third epoch before the transition (epoch -3) had one event per trace, leading to an average count of 1 with no variance. In epoch -2, two traces had no events, and one trace had one event, leading to an average count of one third and a standard error of the mean of plus or minus one third. The calculations were continued as such for the other two epochs, leading to the plot in Figure 2B which allows one to see the dip in event rate leading up to the transition and rise right after. This example is a subset of a larger data set, showing only three ROIs and four epochs. The actual trajectory plots are summaries of all ROIs (and all unique ROI pairs for the correlations) for each animal across ten epochs (nine before the transition and one after).

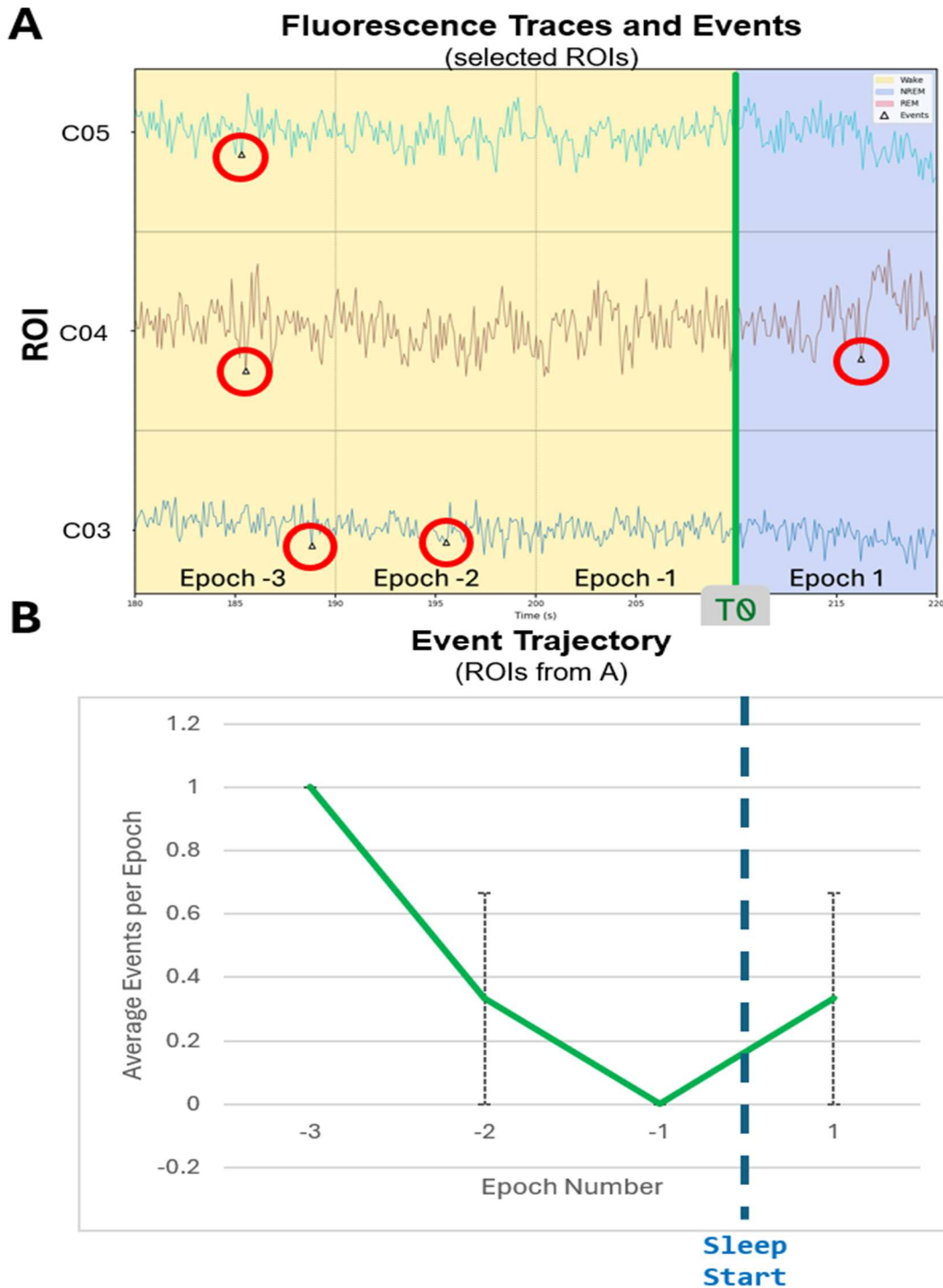


Figure 2: Trajectory Analysis Example

Sample data demonstrating how an event rate trajectory plot with ROIs and three epochs before the transition would be made. **A:** Traces from three ROIs in $\Delta F/F$. Background colors show vigilance state, with yellow being wake and blue being NREM. Event starts (i.e. the troughs before sufficient peaks) marked with white triangles highlighted with red circles. Transition moment (border between epochs in different states) marked with a green line labeled T0. Epoch positions relative to the transition labeled on the bottom trace. **B:** Plot of mean event rate of the three example traces in events per epoch. Error bars represent SEM and T0 as defined in A is represented by a blue dotted line.

Our next goal was to examine the way cell interactions change right before sleep onset. Trajectory analysis was also used to examine pairwise cell activity patterns leading up to sleep state transitions using Pearson's product moment correlation as a metric. The same analysis windows from the event rate analysis were used, but instead of averaging a measure for each trace, correlation was calculated for every pair of traces, averaged and plotted. Negative correlations were retained rather than excluded or converted to absolute value, as either transformation would inflate mean correlation estimates, anticorrelation is a valid mode of cell interaction, and there is no strong justification to do so. This provided us an unbiased estimate of net network synchrony.

To determine whether the distance between ROIs had an effect on correlation, the data points were binned according to the distance between the ROIs and re-plotted. The distance bins were determined by ranking the ROI pairs by distance (in pixels) and separating the ranked list into thirds. A table of bin boundaries for all animals can be found in Appendix 6.

Statistics

Statistics were restricted to NREM delta power and SNR. Due to the challenges in determining normality of such a small sample size, both parametric and non-parametric tests were used. The sleep architecture measures used Friedman tests and repeated measures ANOVAs to compare by condition, with *post hoc* Bonferroni-corrected pairwise Wilcoxon and t-tests to find which exact conditions differed. Wilcoxon and t-tests were used to compare the SNR measures of the viral animal (Male 1) to the other animals combined. Significant results with $p < 0.05$ and the tests through which that significance was determined were reported in the figure captions.

Sample size recommendations for future studies were calculated for the trajectory analysis measures. This was done using power analysis for a paired t-test comparing baseline to sleep deprived day effects. Coefficients of variance within and between animals were also calculated to test the validity of the repeated measures design.

RESULTS

Sleep Architecture Corroborates Adequate Sleep Deprivation

To validate that the animals had normal sleep and response to sleep deprivation, sleep stage percentages and NREM delta power were calculated for the baseline and sleep-deprived days. Since the reporter dynamics and method of reporter insertion are not expected to influence overall sleep patterns, all four animals were analyzed together here and only here. The amount of NREM increased and the amount of wake decreased after sleep deprivation as expected. The amount of REM was also very low for both days, justifying omission of the stage from further analyses. NREM delta power, a sensitive marker of sleep pressure, increased significantly between the days despite an n of only 4, validating that the degree of sleep deprivation was sufficient to cause changes in cellular behavior.

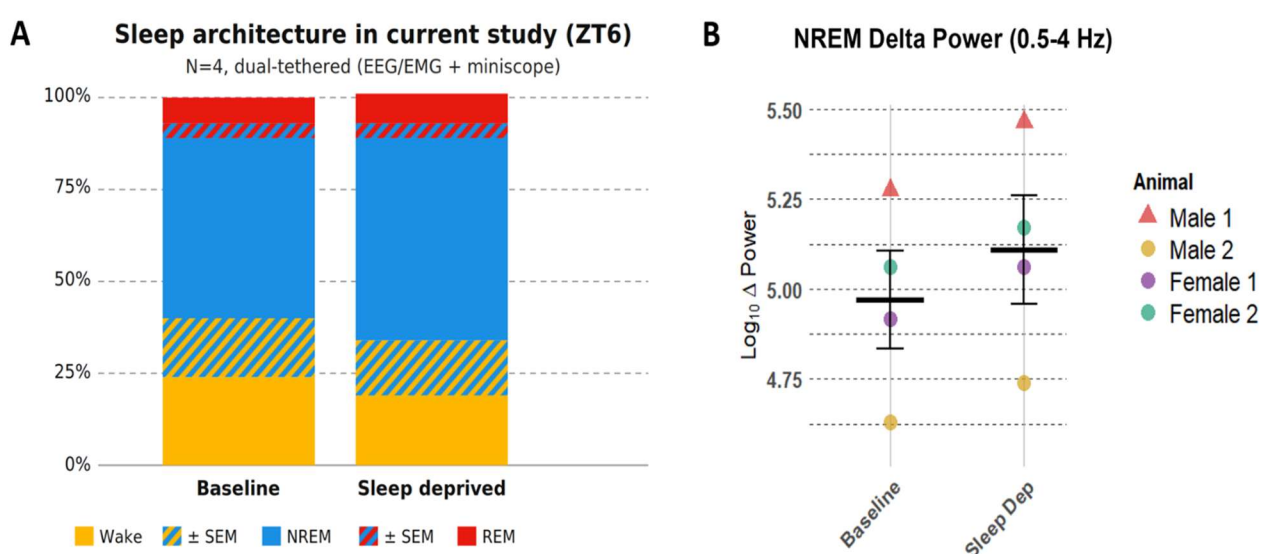


Figure 3: Sleep Architecture

Charts to show the general structure of the animals' sleep over the course of the recording. **A:** Stacked bar charts representing the percentage of the recording each animal spent in each sleep stage. Yellow represents wake, blue represents non-REM, and red represents REM. Hatched areas cover the SEM of the averages for the four animals. **B:** NREM delta power. Each animal's $\log_{10} \Delta$ power is plotted in a different color, with baseline values on the left and values post sleep deprivation on the right. Crossbars represent the mean and error bars represent standard error of the mean. Male 1 is the red triangle, male 2 is the orange circle, female 1 is the purple circle, and female 2 is the green circle. Condition differences are significant according to a paired t-test, with $p = 0.0054$, but not according to a Wilcoxon signed-rank test, with $p > 0.1$.

Signal to Noise Ratio Differs Between Animal Preparations

Since the reporter and transfection methods varied, signal to noise ratio was calculated to see whether Male 1 was similar enough to the other animals to analyze all four together. Figure 4 shows a plot of signal to noise ratio (SNR) for every trace for each animal and plots of its components, mean signal amplitude and baseline noise. Male 1 had both significantly higher signal and significantly higher noise than the other animals. Although they partially canceled out, Male 1 also had a significantly higher signal to noise ratio than the other animals. The difference justifies separating the animals for analysis. Male 2, Female 1, and Female 2 are the focus of this report due to higher n, but Male 1's data are also plotted in Appendix 7.

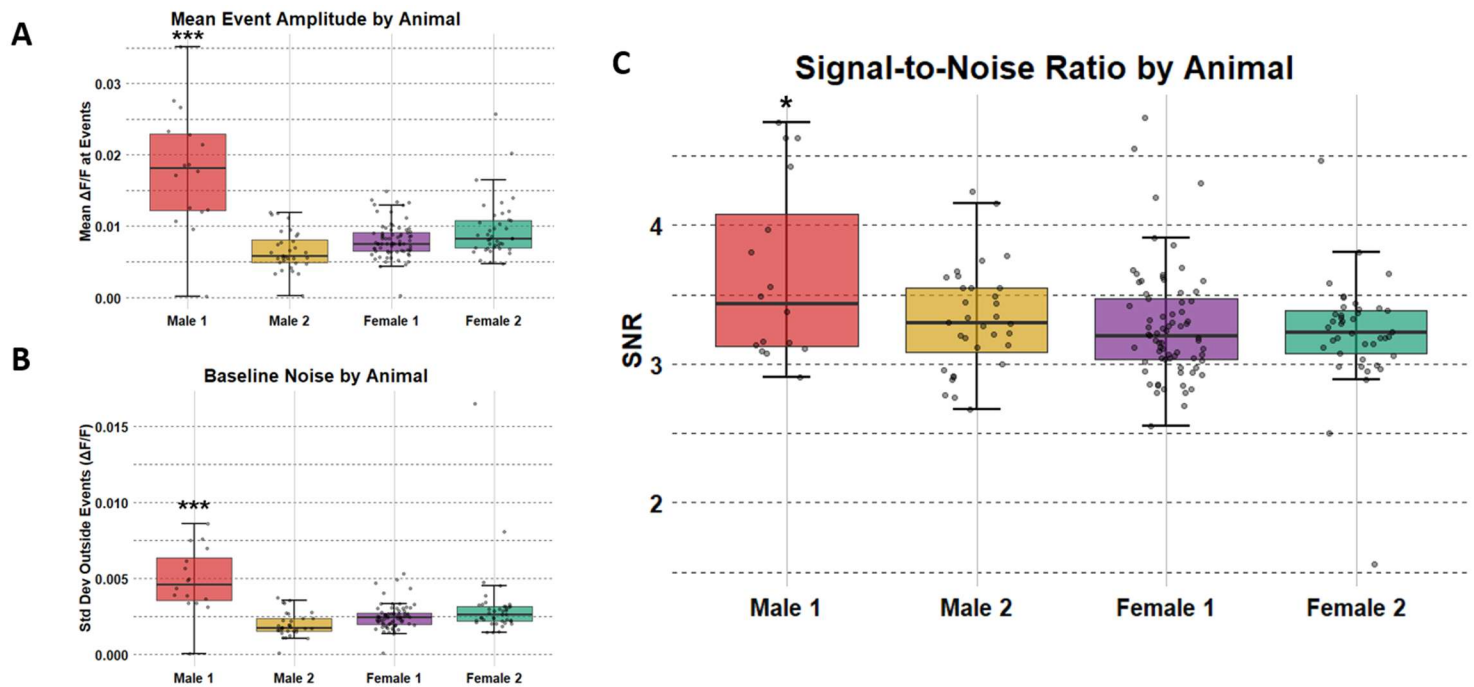


Figure 4: Signal to Noise Calculation

Box plots showing the calculated signal to noise ratio for the $\Delta F/F$ traces and the components thereof. From left to right, each plot shows Male 1 in red, Male 2 in orange, Female 1 in purple, and Female 2 in green. Outliers are defined as any points more than 2.5 times the IQR away from the next most extreme point. **A:** A plot showing the mean event amplitude, in mean $\Delta F/F$ of all parts of all traces within 2 seconds of automatically detected events. The virally transfected animal, Male 1, had a significantly higher signal amplitude than the combined transgenic animals, with $p = 2.8 \times 10^{-7}$ on a Wilcoxon rank-sum test. **B:** A plot of baseline noise, in standard deviation of the portions of each ROI trace (in all three recordings for each animal) more than 2 seconds away from any automatically identified event. The virally transfected animal, Male 1, had a significantly higher noise amplitude than the combined transgenic animals, with $p = 3.2 \times 10^{-7}$ on a Wilcoxon rank-sum test. **C:** Signal-to-Noise ratio, the quotient of mean event amplitude and baseline noise for each trace. The virally transfected animal, Male 1, had a significantly higher SNR than the combined transgenic animals, with $p = 0.039$ on a Wilcoxon rank-sum test.

Event Rate Increases After Sleep Deprivation

Event rate trajectory plots showing nine epochs before transitions and one epoch after are shown in Figure 5. Two of the three animals, Male 2 (Figure 5A) and Female 2 (Figure 5B), show a distinct increase in event rate after sleep deprivation, aligning with our hypothesis. We can see a difference in cell activity on the scale of a day in these animals. They do not show distinct activity trends on the scale of seconds to minutes, despite the expected increase derived from the literature. Female 1 (Figure 5C) opposed all hypotheses, with no distinct trend on the baseline day and a gradual decrease in activity approaching transition after sleep deprivation.

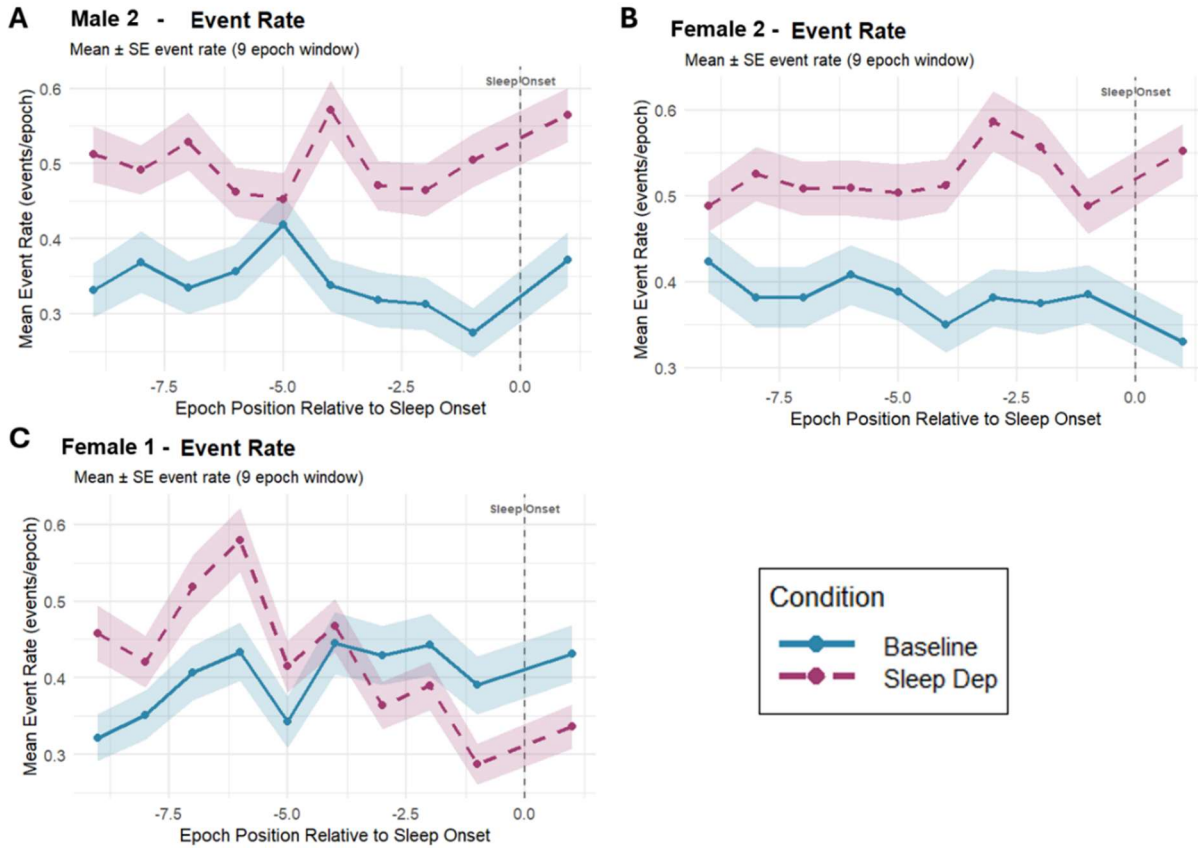


Figure 5: Event Rate Trajectory

Ten-epoch pairwise event rate trajectory plots separated by animal. Blue solid lines show trajectories for baseline days, and magenta dashed lines show trajectories for sleep deprivation days. The point positions show the mean event rate for all wake to NREM transitions in each animal and the error bars represent SEM. The black dotted line represents sleep onset. **A:** Event rate for Male 2. **B:** Event rate for Female 2. **C:** Event rate for Female 1.

Pairwise Correlation Increases After Sleep Deprivation

Pairwise correlation trajectory plots showing nine epochs before transitions and one epoch after are shown in Figure 6. Two of the three animals, Male 2 (Figure 6A) and Female 2 (Figure 6B), show a distinct increase in pairwise correlation after sleep deprivation, again aligning with our hypothesis. This increase is not statistically significant but is consistently far greater than the standard error of the mean. We can see a difference in cell activity on the scale of a day in these animals. They do not show distinct correlation trends on the scale of seconds to minutes despite the expected increase derived from the literature. Female 1 (Figure 6C) opposed all hypotheses, with no distinct trend on either day. While no animals consistently had significant r differences,

Female 1 got the closest because it had many more ROIs leading to orders of magnitude more pairs.

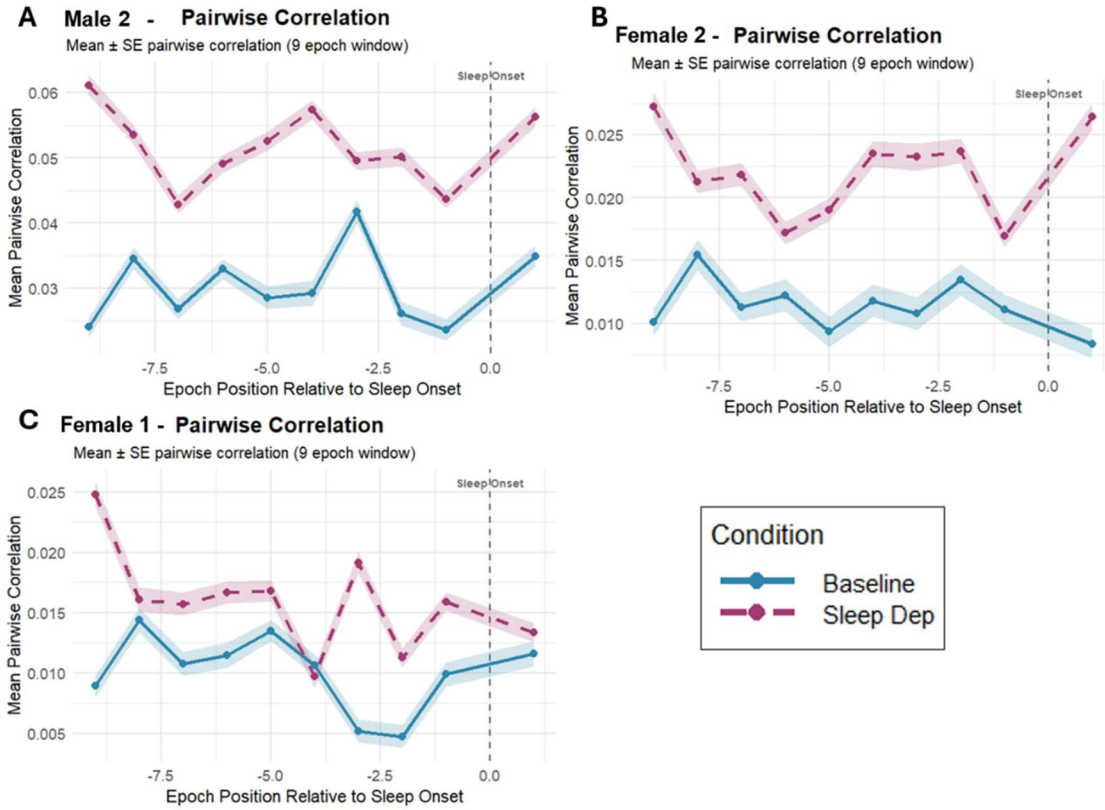


Figure 6: Pairwise Correlation Trajectory

Ten-epoch pairwise correlation trajectory plots separated by animal. Blue solid lines show trajectories for baseline days, and magenta dashed lines show trajectories for sleep deprivation days. The point positions show the mean correlation for all wake to NREM transitions in each animal and the error bars represent SEM. The black dotted line represents sleep onset. **A:** Correlation for Male 2. **B:** Correlation for Female 2. **C:** Correlation for Female 1.

Pairwise Correlations Increase More in Far Cell Pairs Than Near

Distance binned pairwise correlation trajectory plots showing nine epochs before transitions and one epoch after are shown in Figure 7. Two of the three animals, Male 2 (Figure 7A) and Female 2 (Figure 7B), show a distinct increase in pairwise correlation after sleep deprivation, again aligning with our hypothesis. This increase is not statistically significant but is consistently far greater than the standard error of the mean. Surprisingly, this difference is bigger in far cell pairs than in near cell pairs, contradicting our hypothesis about maximally proximal connectivity. We can see a difference in cell activity on the scale of a day in these animals. They

do not show distinct correlation trends on the scale of seconds to minutes despite the expected increase derived from the literature. Female 1 (Figure 7C) opposed all hypotheses, with no distinct trend on either day.

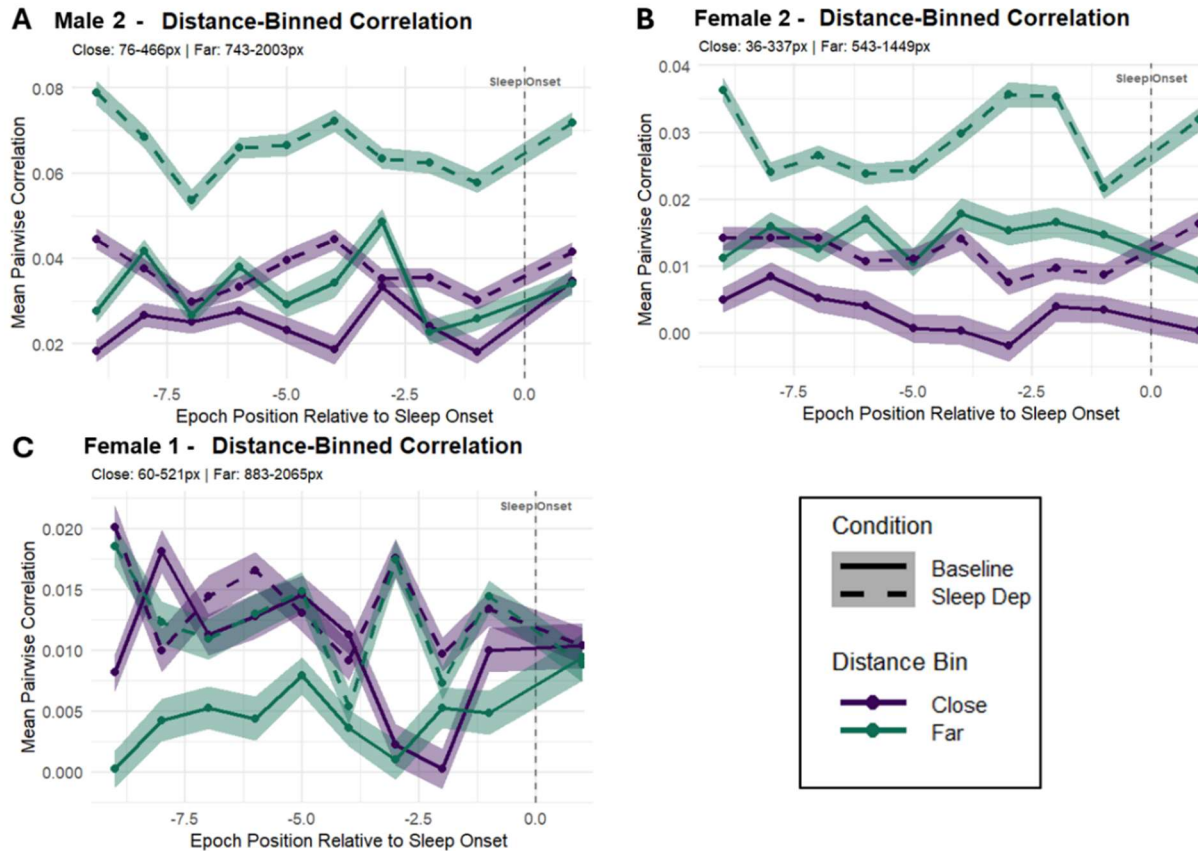


Figure 7: Distance-Binned Pairwise Correlation Trajectory

Ten-epoch distance-binned pairwise correlation trajectory plots separated by animal. Solid lines show trajectories for baseline days, and dashed lines show trajectories for sleep deprivation days. Dark purple lines show close pairs and green lines show far pairs. The point positions show the mean correlation for all wake to NREM transitions in each animal and the error bars represent SEM. The black dotted line represents sleep onset. **A:** Correlation for Male 2. **B:** Correlation for Female 2. **C:** Correlation for Female 1.

Coefficient of Variation Conditionally Higher Between Animals Than Within

Within-animal variability coefficient of variation (CV) was 28% for correlation and CV of 14% for event rate. Between animal variability was dependent on the measure, with correlation CV being only 67% and event rate CV being 5%. All coefficients of variation are presented in Appendix 9. The substantially larger between-animal variability in correlation, relative to within-

animal variability, suggests that baseline correlation differs considerably across individuals, supporting the use of a repeated-measures design.

Power Analysis Suggests Larger Sample Size Needed

We did a power analysis to determine how many animals would be needed to detect a 30% difference, with the simplifying assumption that we only need to find differences between the baseline and sleep deprivation day using a paired t-test. The results are shown in Figure 8. Event rate required the fewest animals at 4, while pairwise correlation required 24. If correlation was further split by distance, differences in the close and medium pairs could be found with only 16 animals while 60 animals would be needed to detect differences in the far pairs.

The variance was much higher between animals than within animals, supporting the use of repeated measures designs such as this for future studies. Specific coefficients of variation for every animal and condition can be found in Appendix 9.

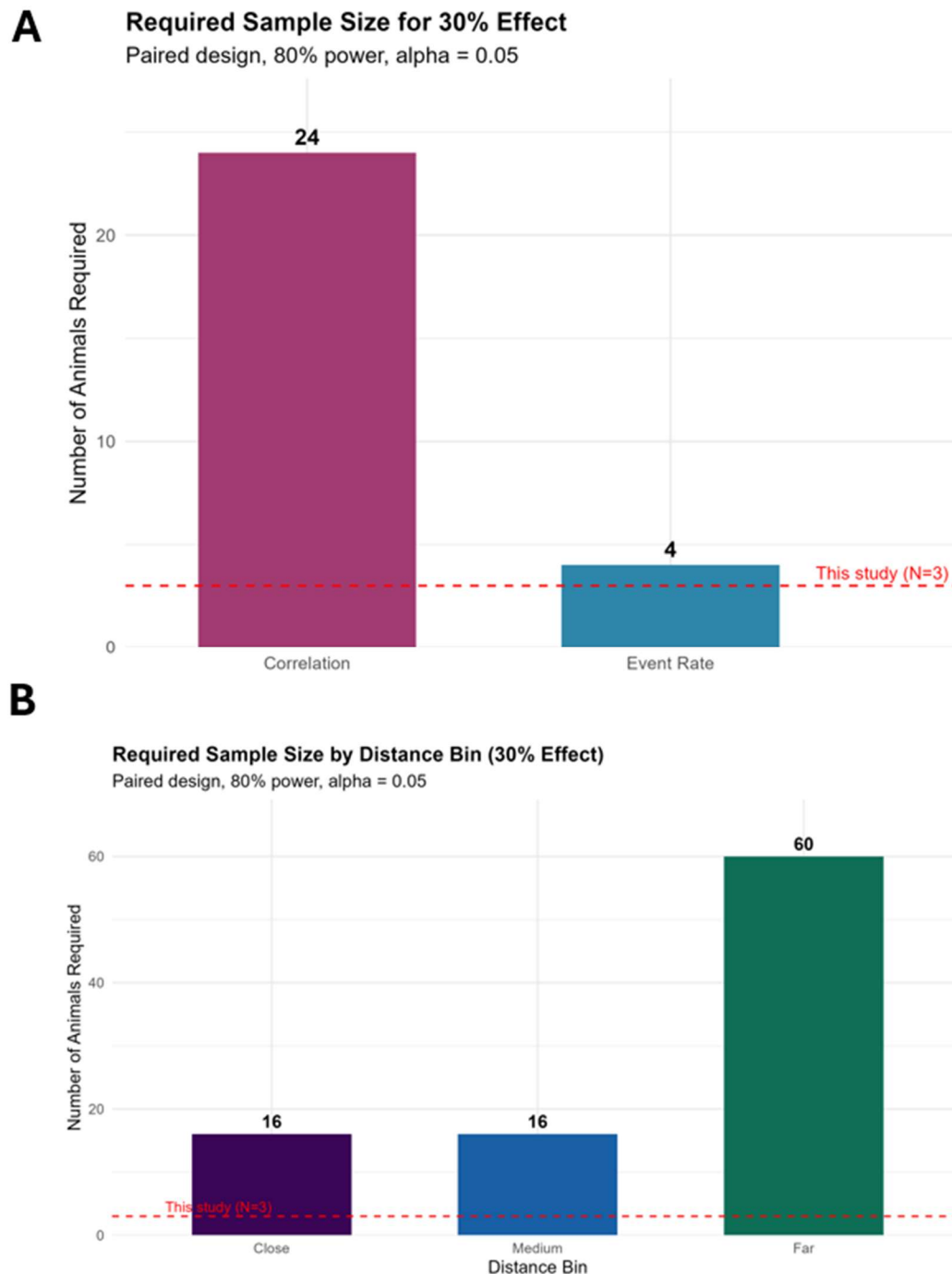


Figure 8: Power Analysis

Sample size required to detect a 30% effect, calculated using a paired t-test power analysis based on the effect size of sleep deprivation. The red dotted line represents the maximum sample size of this study, which is three animals with the same methods. **A:** Sample size required to detect differences in correlation coefficient on the left and event rate on the right. **B:** Sample size required to detect differences in correlation further separated by the distance between each pair, with close, medium, and far pairs shown left to right.

DISCUSSION

Sleep Deprivation Elevates Neural Activity and Pairwise Correlations in mPFC

Two of the three animals showed a consistent increase in both event rate and pairwise correlation, aligning with the prediction that elevated sleep pressure would increase neuronal activity. This is consistent with findings in the barrel cortex, where cortical firing rates increase systematically with the duration of prior wakefulness and decrease with sleep, a pattern tightly correlated with NREM slow wave activity (11). The effect size was approximately 0.1 events per epoch, which was modest in absolute terms but distinguishable from baseline given the tight standard error across the pre-transition window. A similar pattern held true for pairwise correlation, with sleep-deprived animals showing consistently higher mean correlation than baseline days in the same animals. The fact that both measures moved in the same direction in the same animals in line with predictions lends some credibility to the finding despite the small sample size. The third transgenic animal, Female 1, did not replicate this pattern and proposed reasons for this are discussed in the section on individual variability below.

It should be noted that these condition-level differences were not tested for statistical significance, since the sample size of three animals was insufficient to power such comparisons. The observed separation between baseline and sleep-deprived trajectories is presented as a descriptive trend consistent with literature, not a confirmatory result.

Pre-Transition Null Result Has Technical and Biological Candidate Explanations

No systemic change was observed across the 90-second window preceding sleep onset in any animal, meaning neither event rate nor pairwise correlation showed the expected increase in NREM relative to wake on the micro scale. The expected pre-transition increase in firing derives from the logic that slow wave activity builds before sleep onset and is reflected in increased neuronal synchrony and firing (11). Simply put, the macro-scale condition effect was recoverable under these constraints while the micro-scale temporal dynamics were not.

There are two classes of explanation for this deviation from expectations: technical and biological. One technical explanation is limitations of the recording paradigm, particularly the 10 Hz sampling rate and lack of deconvolution. These together limit sensitivity to the sub-second dynamics that characterize the transitions between “on” and “off” periods that underlie slow wave generation (14).

Additional technical explanations involve temporal parameters of the analysis. The 10-second epoch length may have obscured finer dynamics. Slow oscillations build on a timescale of seconds, and a 10-second bin averages across multiple “on” and “off” cycles, potentially washing out directional trends that would be visible at higher temporal resolution. Smaller epochs would trade statistical stability for temporal precision, which may be the right tradeoff for capturing transition dynamics specifically. It should be noted, though, that epoch length has a practical floor, as scoring burden scales with both the difficulty of and the number of individual scoring decisions. Epoch lengths under five or so seconds may render manual scoring at typical recording lengths impractical. The choice to examine nine epochs before transition and only one after also warrants reconsideration. Restricting the post-transition window to a single epoch was done under the assumption that the most informative dynamics were in the pre-transition period, but changes in event rate and correlation may continue to evolve in the epochs immediately following sleep onset. These changes could inform whether the null result reflects a genuine absence of pre-transition build-up or simply a gradual rather than anticipatory change.

Our final candidate explanation is biological. The mPFC may not exhibit the same pre-transition dynamics as regions where such patterns have been characterized, not because the region is unresponsive to sleep pressure, but because its role in sleep differs. As discussed in the Introduction, prefrontal regions show high convergence and divergence of slow oscillation propagation (31,32), which may mean that the mPFC receives and coordinates oscillatory input rather than generating it locally in a manner that would produce a gradual pre-transition build-up.

These explanations are not mutually exclusive, and a follow-up study varying epoch and window parameters would be the most direct way to distinguish between them.

Far-Pair Correlations Show Unexpected Sensitivity to Sleep Deprivation

The most novel finding of this study is that the increase in pairwise correlation after sleep deprivation was larger in far cell pairs than in close cell pairs, at least in the two animals showing consistent sleep deprivation effects. This directly contradicts the prediction that nearby pyramidal cells, more likely to share direct synaptic connections, would show stronger coordination due to mutual excitation. A few non-exclusive interpretations are worth considering.

First, the far pairs themselves may be the ones with direct synaptic connections. Pyramidal cells in the mPFC are known to form long-range intracortical connections, and it is possible that the pairs captured by the far distance bin are synaptically coupled in ways that close pairs are not.

Second, close pairs may already be near the correlation ceiling at baseline due to dense local connectivity, leaving less dynamic range to express an increase driven by sleep deprivation. Far pairs, operating at lower baseline correlation, may be more sensitive to the effect.

Third, and more speculatively, the pattern may reflect the influence of long-range synchronizing inputs from outside the field of view. If the mPFC receives distributed oscillatory drive, then distant cells sharing afferents from the same region could synchronize more strongly than cells connected only through dense but spatially confined local circuits. This would be consistent with the proposed role of the mPFC as a convergence zone in the anterior sleep network (31,32). This interpretation is also broadly consistent with the framework of local sleep, in which cortical coordination can emerge through mechanisms not strictly dependent on proximity (17-19). This finding is preliminary given the sample size, but it represents the clearest candidate for a novel contribution from this dataset. A follow-up study is warranted and should include a larger sample size and *post hoc* cell type identification to determine whether the far-pair effect is selective to pyramidal cells or reflects a mixed population response.

Individual Variability Reflects Biological and Technical Heterogeneity

Two of the four animals presented here were different than any other animal. Male 1 was a technical outlier due to different reporter version and transfection method. Female 1 underwent the same preparation as Male 2 and Female 2, but its trajectory patterns didn't closely resemble those of the other animals.

The differences in Female 1 and the other animals are of ambiguous origin. There is a chance that the lens of that animal ended up in a different location from the other two, due to irregularities occurring either during or after surgery. Histology (Appendix 2) does not show any huge differences in lens placement, but tissue damage from histology preparation makes it difficult to determine the precise location of the lens tract in any animal. There is also a possibility that multiple response phenotypes exist and Female 1 represents a different phenotype than Male 2 and Female 2. This cannot be determined definitively either way with a sample size of 3. The animal could also be a biological outlier generally. Regardless of the explanation, the implication of these results is that more research is needed. A larger sample size could provide more examples of multiple phenotypes if they exist or provide documentation of the effects of methodological irregularities.

The differences in Male 1 animals were due to a different preparation. This animal had a different reporter and transfection method than the other animals shown here because the precise methods evolved over the course of the project. This animal and others that were not described in detail provide a means of analyzing appropriate methods for various similar studies.

Methodological Considerations Inform Future Calcium Imaging Design

Trace-level signal to noise ratio (SNR) averaged 4.4 across all ROIs, consistent with validated thresholds for event detection in 1-photon calcium indicator with transgenic indicator expression (38,39). While literature indicates that the current SNR is sufficient, a higher SNR makes ROI detection, especially manual detection, easier and more reliable. The dual confounds of not having any of the animals share a reporter but not a transfection method and vice versa prevents us from saying definitively which changes led to the differences, but some clues exist. The peak signal difference of the reporters after several action potentials (the expected behavior regime in our target cells) points towards GCaMP6f being a clearer reporter (40). There is literature,

however, indicating that viral animals have a clearer signal than transgenic animals in 1-photon imaging (39), and the spatial specificity lends credence to that theory. While signal quality is important, there are other factors that are more important in deciding reporter and transfection method.

The use of transgenic animals allowed the exclusion of certain surgical procedures, making the process around 30% faster (Figure 10). This resulted in both higher throughput as it was easier to perform several surgeries a day, and lower time under anesthesia which enabled faster recovery of the animals. The transgenic animals, though, were harder to record from for several reasons. Firstly, there was no regional localization of expression, meaning CaMKIIa-positive cells throughout the brain expressed the reporter. This could have led to fluorescent signals from outside the imaging plane contaminating the signal, a problem that cannot be mitigated at the time of recording using one-photon epifluorescent microscopes such as our miniscope. The viral gene transduction also led to more stable expression of the reporter. We were able to record from our virally transduced mouse seven months after the injection and implantation surgery, while the signal from the dual transgenic mice degraded to levels that couldn't reliably be detected after around five months. We know this because there were two animals that successfully recovered from the surgery but did not undergo recording and sleep deprivation until after the reporter had degraded, leading to an undetectable signal. Finally, the two methods yielded different levels of signal and noise, which partially canceled each other out and resulted in less stark differences in signal to noise ratio, but nonetheless yielded different enough recordings that separating the animals by transduction type was more appropriate for statistical analysis. The higher signal and noise in the virally transduced animal implied that it had higher background contamination. Future studies should carefully consider the tradeoffs of the different methods. The gains in surgical simplicity and consequent success rate should lead many to adopt the transgenic method for animal welfare reasons, but those requiring specific spatial targeting or extended recording spans may opt to use viral vectors instead.

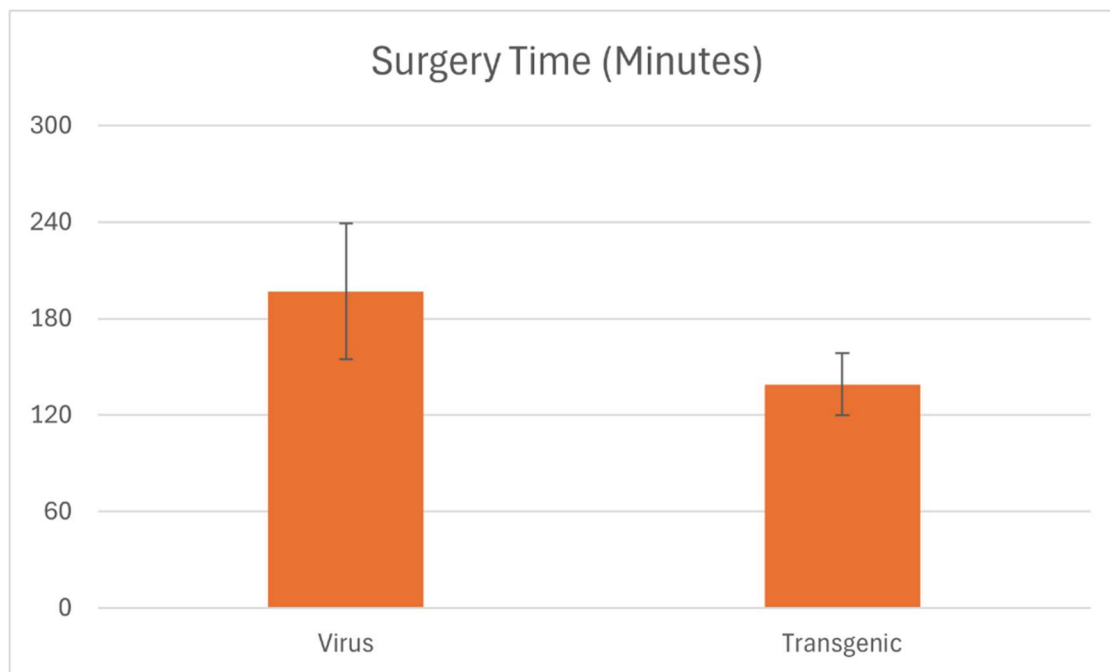


Figure 9: Surgery Durations

Mean durations of seven surgeries of each type, defined as the amount of time between the animal being put on anesthesia and the animal being taken off anesthesia. Error bars represent standard deviation.

Due to the length of recording and resultant photobleaching and data volume, we only recorded at 10 frames per second. This meant that any temporal resolution gains couldn't be captured by our system (40). These factors, along with the fact that all of Inscopix's standard recording and analysis pipeline is optimized for GCaMP6, leads us to the conclusion that GCaMP6f would be the best reporter for similar future experiments.

Automated ROI selection could help with some of the above challenges in future studies. Manual ROI selection may be familiar and seem simple to many, but automated ROI selection allows advanced signal decontamination techniques that we suspect to be worth the trouble of setting up an automated ROI detection pipeline. CNMFe would have mitigated some of the signal to noise ratio issues that we had (41,42). Those looking to conduct similar studies should use the CNMFe functionality built into IDPS or attempt to utilize the open-source CalmAn Python package (43) if adept with such software (since it has a steep learning curve).

The medial prefrontal cortex may not be the optimal target for this type of analysis. While it does mediate many processes that are disrupted by sleep deprivation, the direct effects may be more apparent in other regions or longer time scales.

Pyramidal cells also may not be the optimal target. Recent literature has highlighted the role of CAMKII-expressing parvalbumin interneurons in response to sleep deprivation (44). This calls into question not only the idea of targeting pyramidal cells in general, but also the actual specificity of the reporter. The finding that CAMKII (our CRE driver in both methods) is expressed in parvalbumin cells in response to sleep deprivation implies the possibility that some of the dynamic puncta may be parvalbumin cells instead of pyramidal cells. The presence of such cells would serve as a parsimonious explanation for the presence of REM-active cells in the field of view, which are presented in Appendix 8. Regardless of optimal targeting, this uncertainty highlights the need for *post hoc* cell identification in similar studies, which can be achieved by light-guided sectioning (45).

Limitations & Future Directions

The most crucial limitation of this study is the small sample size. This was mostly influenced by surgical complications that could be mitigated in the future by using transgenic transfection and recording early enough to avoid reporter degradation.

There are several technical limitations of this study as well. The use of 1-photon imaging limits our ability to spatially isolate our signal, and that could be mitigated by the use of 2-photon miniature microscopy that has become commercially available since this project's inception. Due to data volume and photobleaching concerns (the latter of which could also be mitigated by using 2-photon microscopy), we only recorded at 10 frames per second. This, combined with the lack of deconvolution, prevents us from seeing anywhere near the firing rate seen in MUA (2-5 Hz during wake and 0.5-2 Hz during NREM (46)). As a sidenote, the pairwise correlations, despite their small magnitude, were comparable to those found in literature (47-49), highlighting the robustness of the measure.

The experimental scope is also limited. We only examined acute SD in a single brain region in genetically homogenous young adult mice.

Studies such as this have the potential to be fertile grounds for discovery. Even the data set used here hasn't been fully examined. It could be used to examine NREM to wake transitions and recovery day, medium distance or REM sleep phenomena. The recording window could also be optimized to find effects directly adjacent to transitions, with the recording time, transition

windows and epoch sizes available as adjustable parameters. One could also repeat the entire recording paradigm to glean insights into longer term effects. Since these methods utilize freely moving animals, behavioral tests such as 5CRTT (50) could be added to the battery of recordings presented here, and would make for a nice reward-driven decision making parallel (testing the mPFC) to the motor task experiment (testing the motor cortex) that inspired elements of this study (17). Similar studies could also be used to examine the differential responses of brain regions to sleep deprivation but note that we attempted to examine the anterior cingulate area already and its proximity to the superior sagittal sinus rendered lens implantation challenging. Studies of alternate cell types would be warranted, especially considering the higher vulnerability of non-pyramidal cells to sleep deprivation (51) and multifaceted behaviors of various cell types during normal sleep (52). Opto- and chemogenetic techniques offer intricate control of neural activation timing by cell type and optogenetic activation lasers are even conveniently built into some Inscopix miniscopes. Lastly, there are analytical techniques that could be applied to extract different types of information from the same data files. In particular, NMF could be used to pick out cell ensembles that activate under certain conditions (53).

CONCLUSIONS

This study shows that simultaneous *in vivo* fluorescent calcium recording and EEG recording is possible and interpretable. Specific methods have pros and cons that have been outlined here. This technique is good for examining cell behavior across multiple timescales and is well suited for studying sleep.

The data showed a tentative increase in fluorescent event rate after sleep deprivation consistent with literature but did not show the expected increase in NREM sleep compared to wake. They also showed an expected increase in mean pairwise correlation after sleep deprivation but again not the expected increase in NREM compared to wake. The pairwise correlation increase was unexpectedly larger in farther pairs compared to closer pairs, highlighting the need for further research to both examine the newly seen effect and confirm the tentative findings.

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APPENDICES

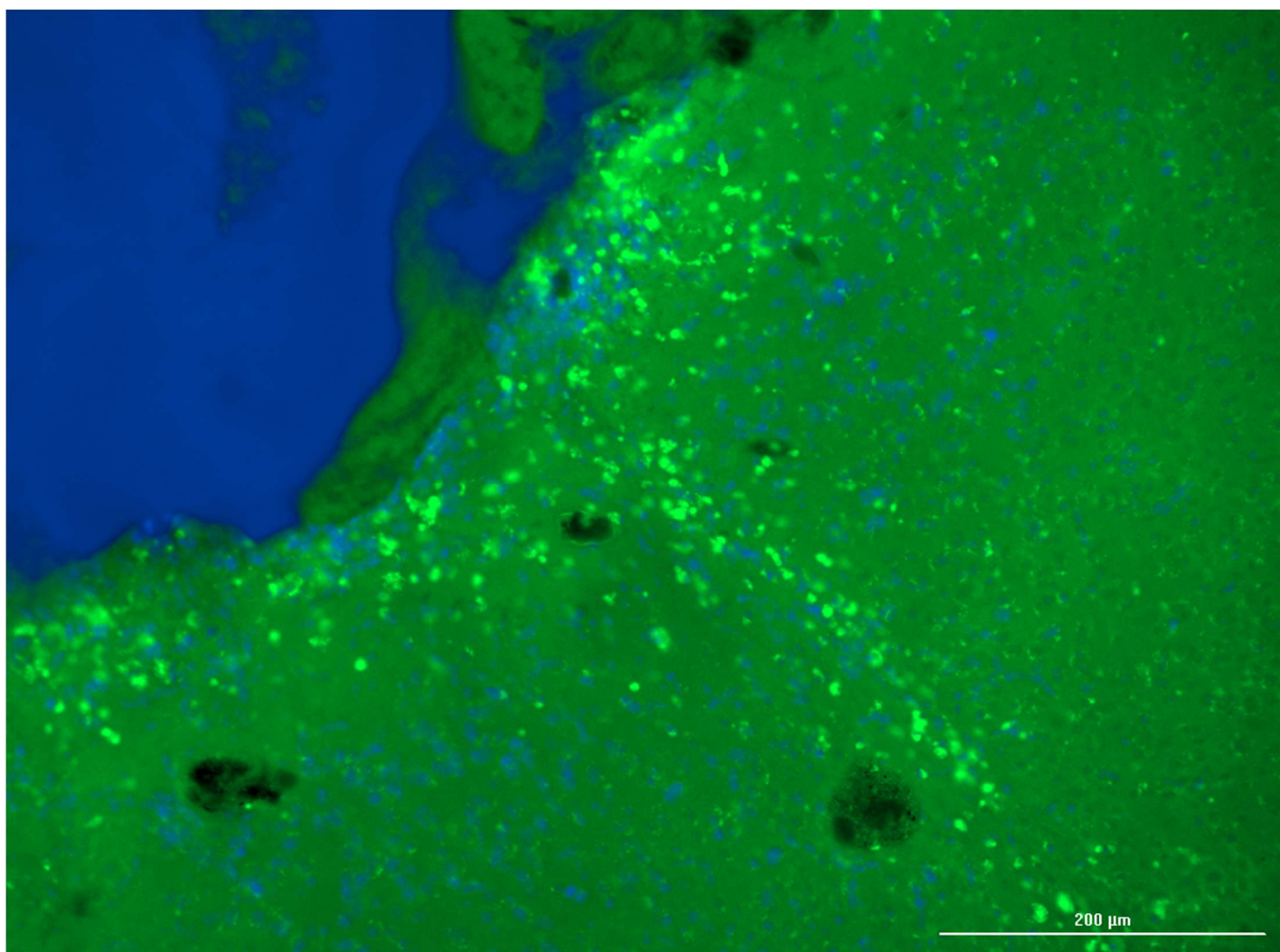
Appendix 1: Materials Table

Item	Manufacturer	Location	Model	Item # (if applicable)
Stereotaxic Alignment System	David Kopf Instruments	Tujunga, CA	1900	
Centering Scope	David Kopf Instruments	Tujunga, CA	1915 40X	
Stereotaxic Alignment Indicator	David Kopf Instruments	Tujunga, CA	1905	
Screw with Wire Leads	Pinnacle Technology	Lawrence, KS	0.10"	8403
Stereotaxic Drill	David Kopf Instruments	Tujunga, CA	1911	
Trephine Bit	Fine Science Tools	Foster City, CA	18004	18004-18
RTI Viral Vector	Bruker Inscopix	Mountain View, CA	Mouse Neocortex AAV1.CaMK2a.GCaMP6f.WPRE.bGHpA	1000-002613
Neuro Syringe	Hamilton	Reno, NV	1µL	80100
Syringe Pump	World Precision Instruments	Sarasota, FL	Micro-2T	
Micropositioner	David Kopf Instruments	Tujunga, CA	2660	
GRIN Lens	Bruker Inscopix	Mountain View, CA	ProView Integrated Prism Lens	1050-004419
Dental Acrylic	Parkell	Edgewood, NY	C&B-Metabond	S380

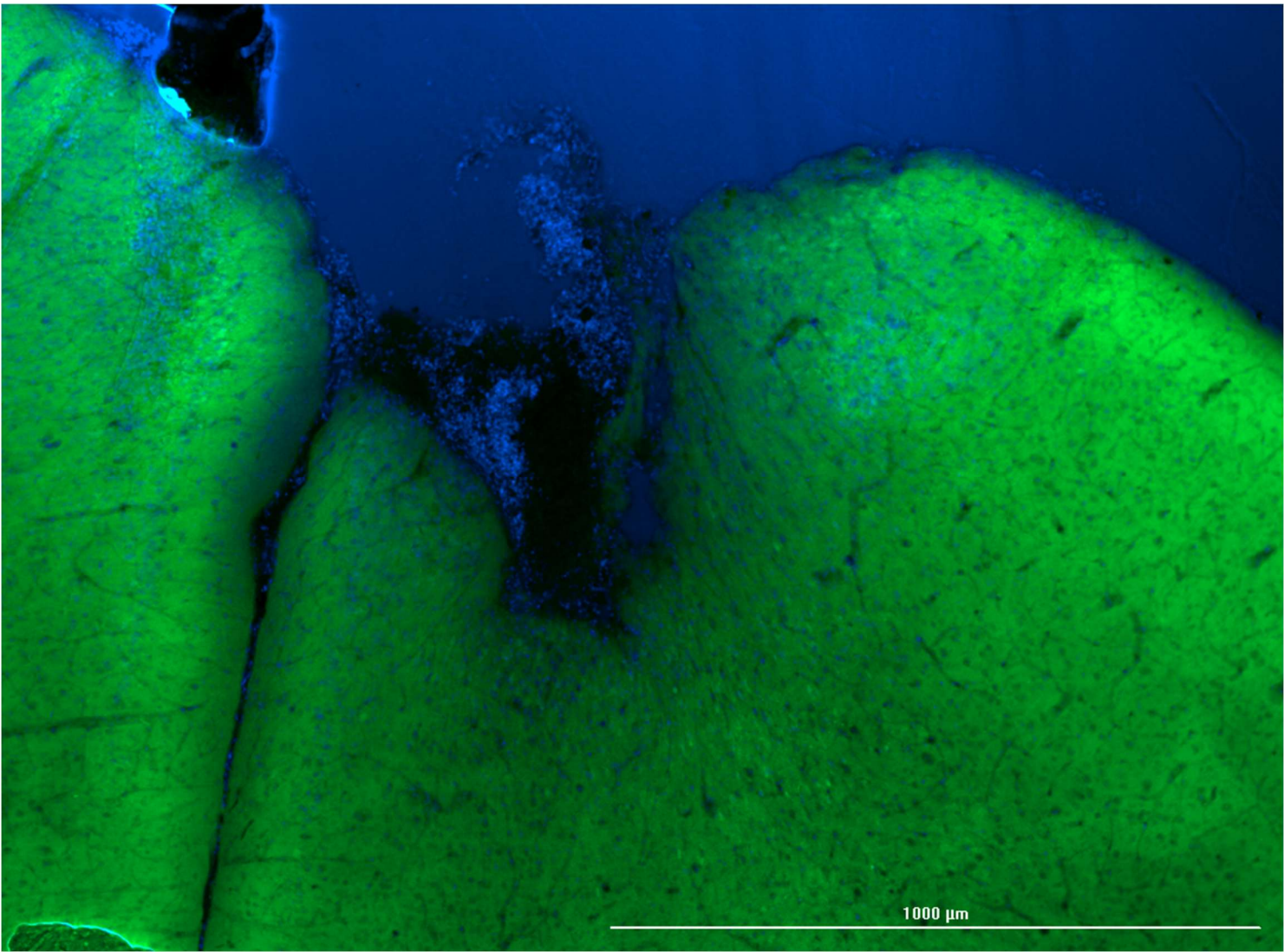
Miniscope and DAQ	Bruker Inscopix	Mountain View, CA	nVoke	100-004245
EEG Preamplifier	Pinnacle Technology	Lawrence, KS	4-Channel Mouse	8406
EEG DAQ	Pinnacle Technology	Lawrence, KS	HR	8206

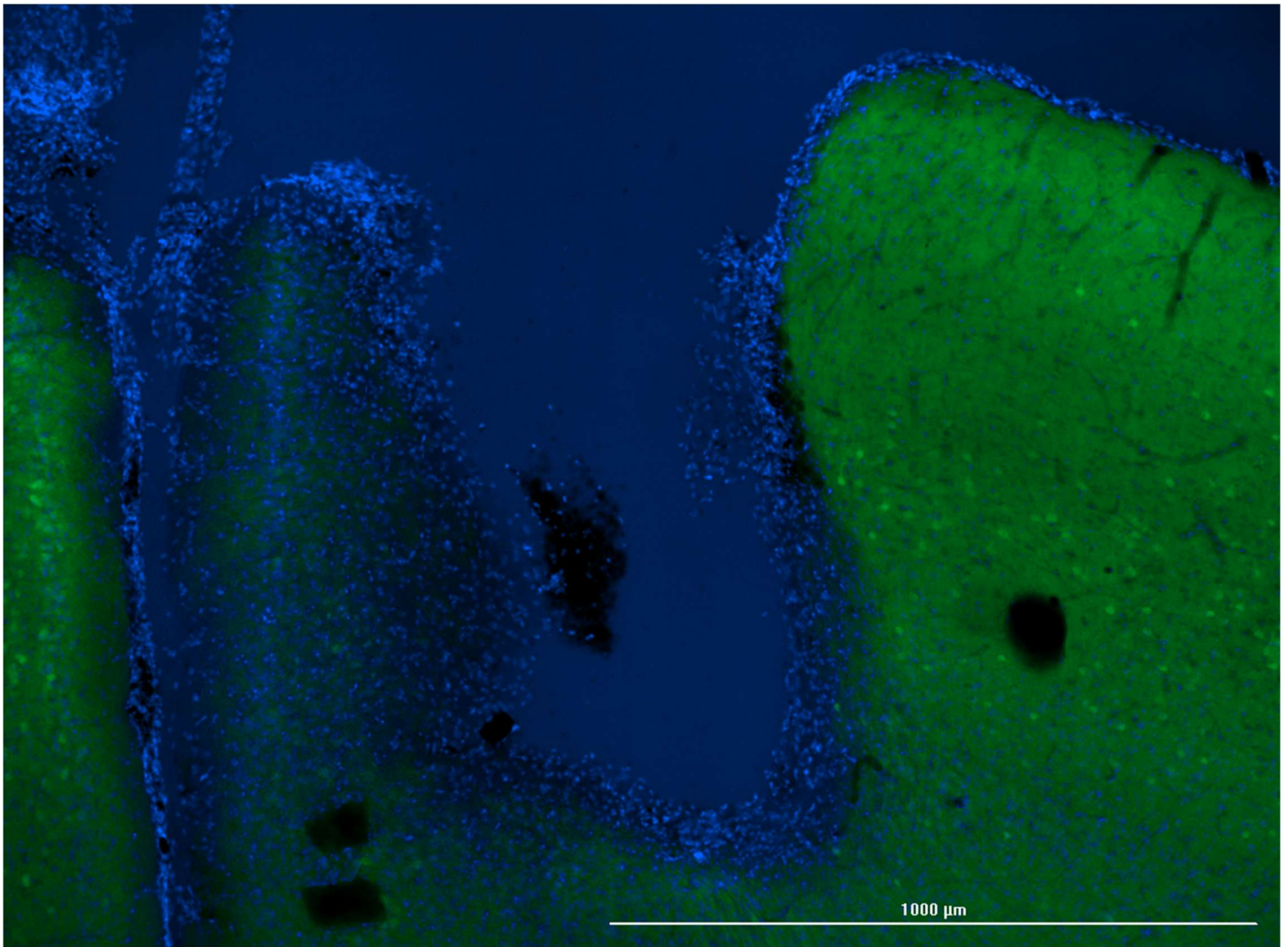
Appendix 2: Histology

Male 1, 10x, GFP and DAPI composite



Female 1, 4x, GFP and DAPI composite

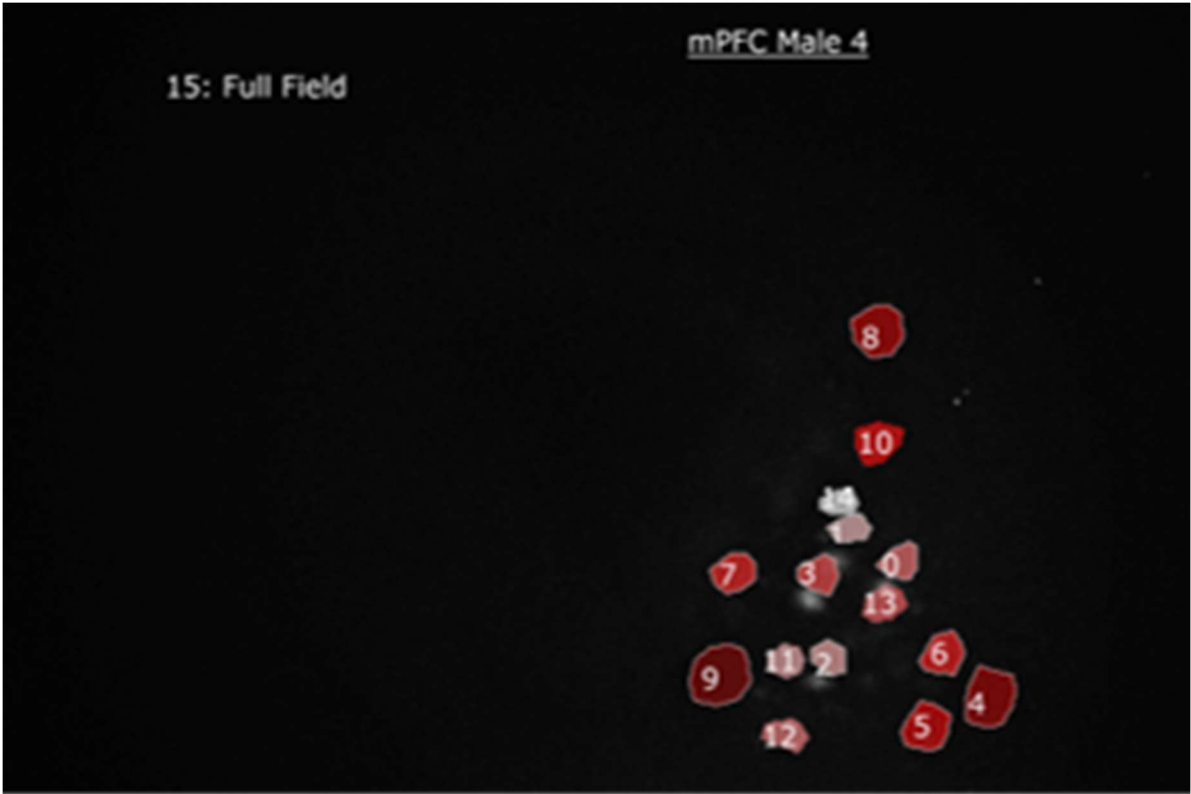




Appendix 3: ROI Maps

All maps are labeled with animal IDs and ROI IDs in white. ROI areas are colorized versions of black and white TIFFs exported by IDPS. The shades of red distinguish different ROIs and nothing more. Backgrounds are gamma-adjusted maximum projects for each animal's baseline.

Male 1



Male 2



Female 1



Female 2



Appendix 4: ROI Filtering Report

Filtering removed any ROIs with an average of less than one automatically detected event per 18 epochs (or 20 events per recording) in the animal's baseline day. The inactive ROIs were excluded from every subsequent analysis.

Animal	Total ROIs	Active ROIs	Percent Retained
mPFCf5	75	73	97.3
mPFCf6	46	41	89.1
mPFCm4	16	16	100
mPFCm9	34	32	94.1

Appendix 5: Window Type Counts

Counts of all windows with 9 consecutive initial state epoch leading up to the transition. These windows were used in every trajectory analysis.

Name	ID	Day	Wake to NREM Count	NREM to Wake Count
Female 1	mPFCf5	Baseline	6	2
	mPFCf5	Sleep Dep	7	2
	mPFCf5	Recovery	11	11
Female 2	mPFCf6	Baseline	11	8
	mPFCf6	Sleep Dep	17	11
	mPFCf6	Recovery	20	17
Male 1	mPFCm4	Baseline	12	6
	mPFCm4	Sleep Dep	10	8
	mPFCm4	Recovery	13	9
Male 2	mPFCm9	Baseline	10	8
	mPFCm9	Sleep Dep	15	10
	mPFCm9	Recovery	8	6

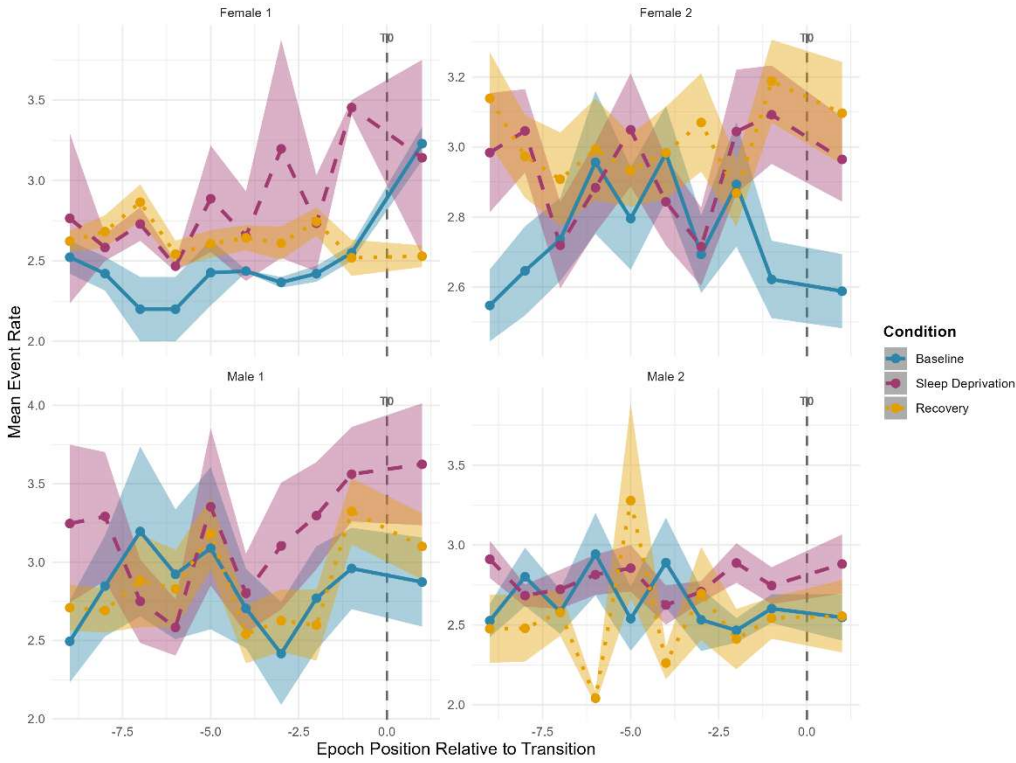
Appendix 6: Distance Bin Boundaries

Animal	Close	Medium	Far
Male 1	64-281px	306-474px	476-1165px
Male 2	76-466px	466-741px	743-2003px
Female 1	60-521px	521-881px	883-2065px
Female 2	36-337px	337-542px	543-1449px

Appendix 7: Graphs of All Animals, Both Transition Types

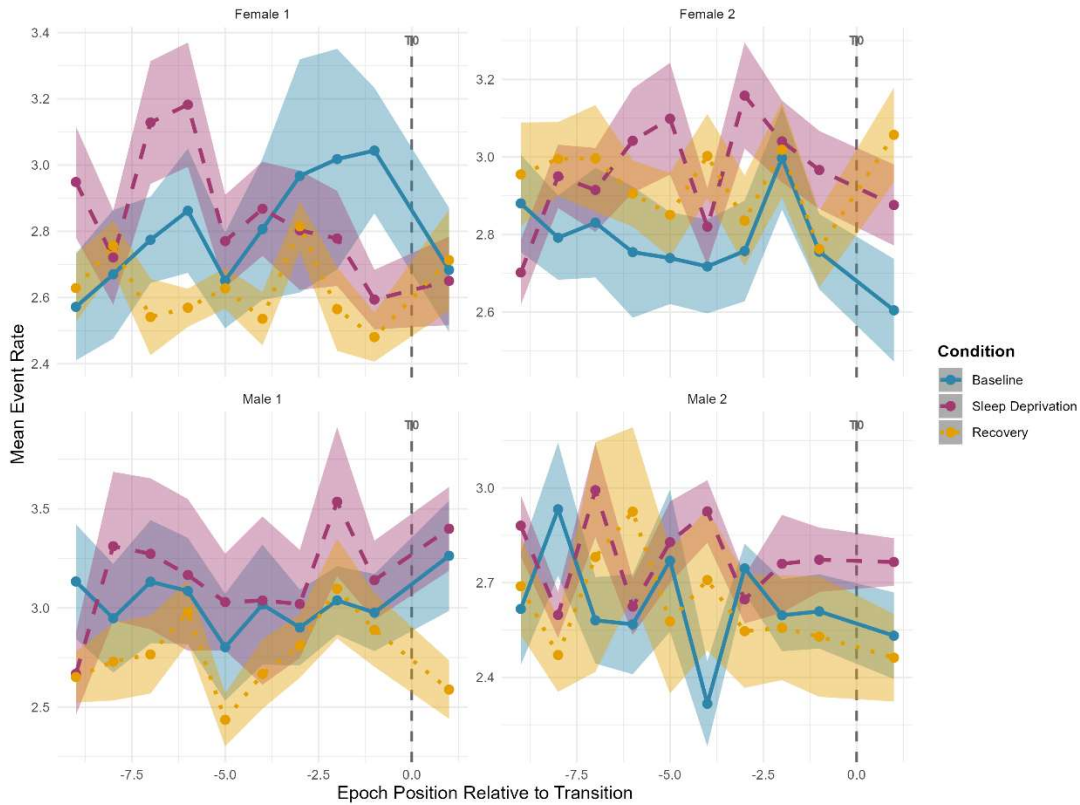
Event Rate Trajectory Across Animals (Active ROIs Only) - NREM2Wake - By Animal

Mean $\pm$ SE events per ROI per transition (9 epoch window)



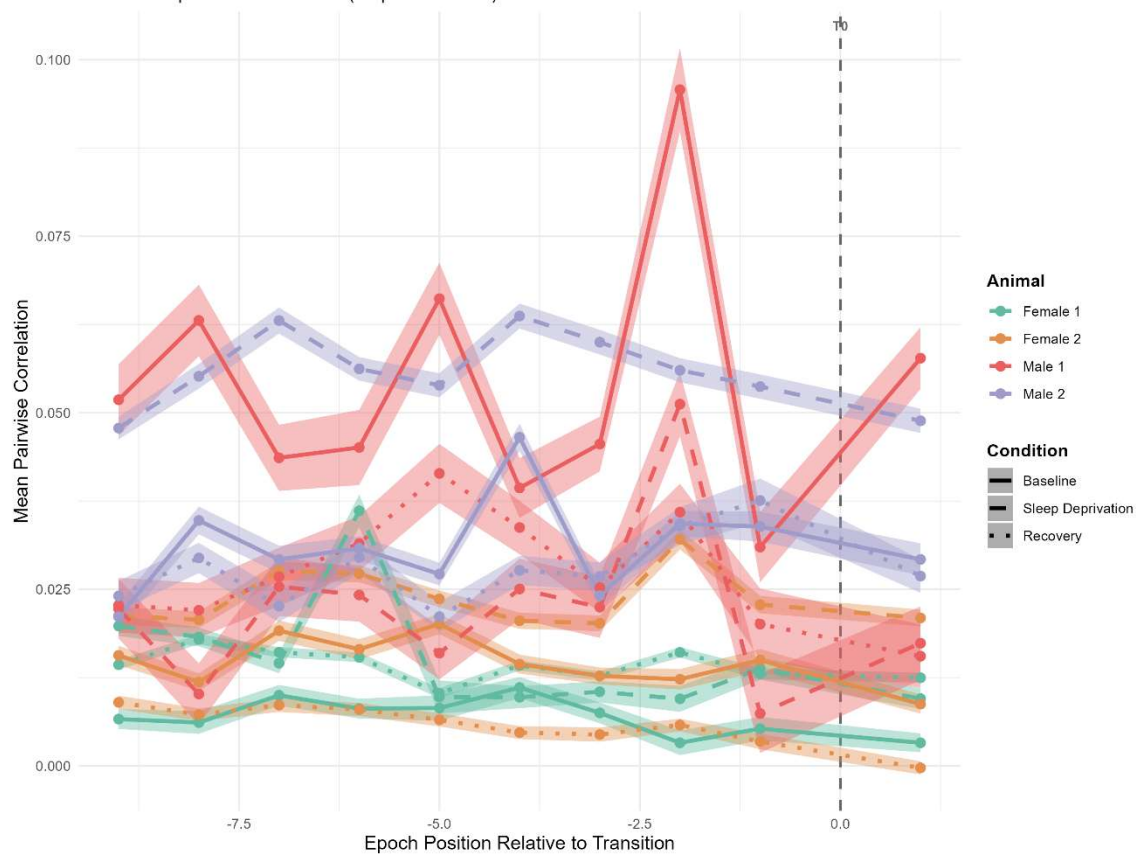
Event Rate Trajectory Across Animals (Active ROIs Only) - Wake2NREM - By Animal

Mean $\pm$ SE events per ROI per transition (9 epoch window)



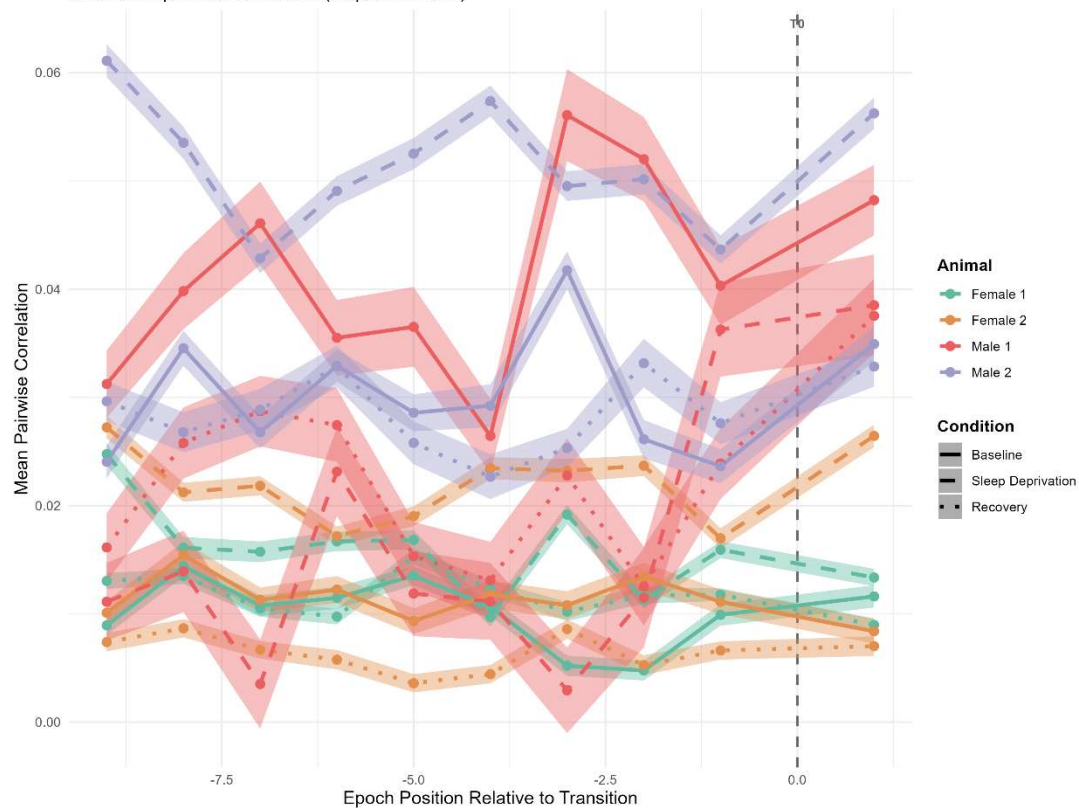
Correlation Trajectory Across Animals (Active ROIs Only) - NREM2Wake

Mean $\pm$ SE pairwise correlation (9 epoch window)

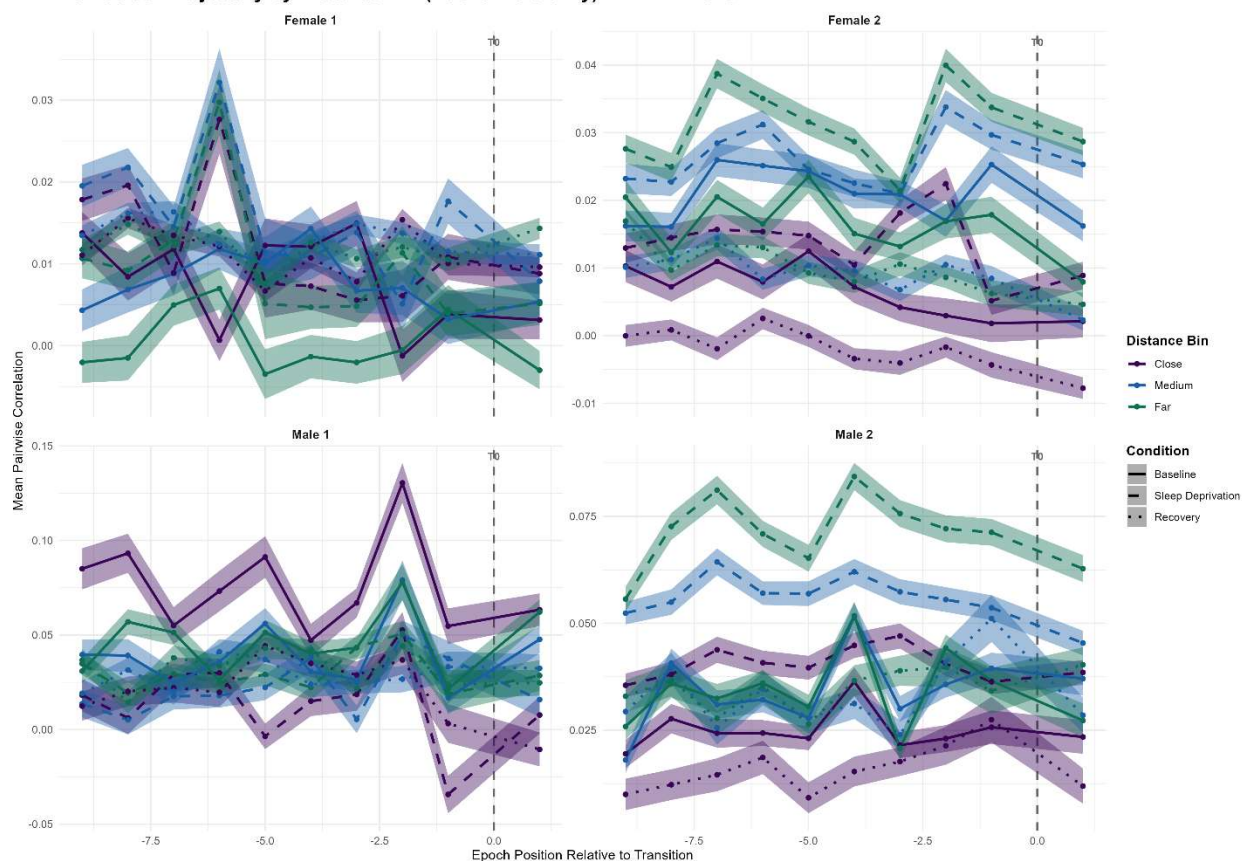


Correlation Trajectory Across Animals (Active ROIs Only) - Wake2NREM

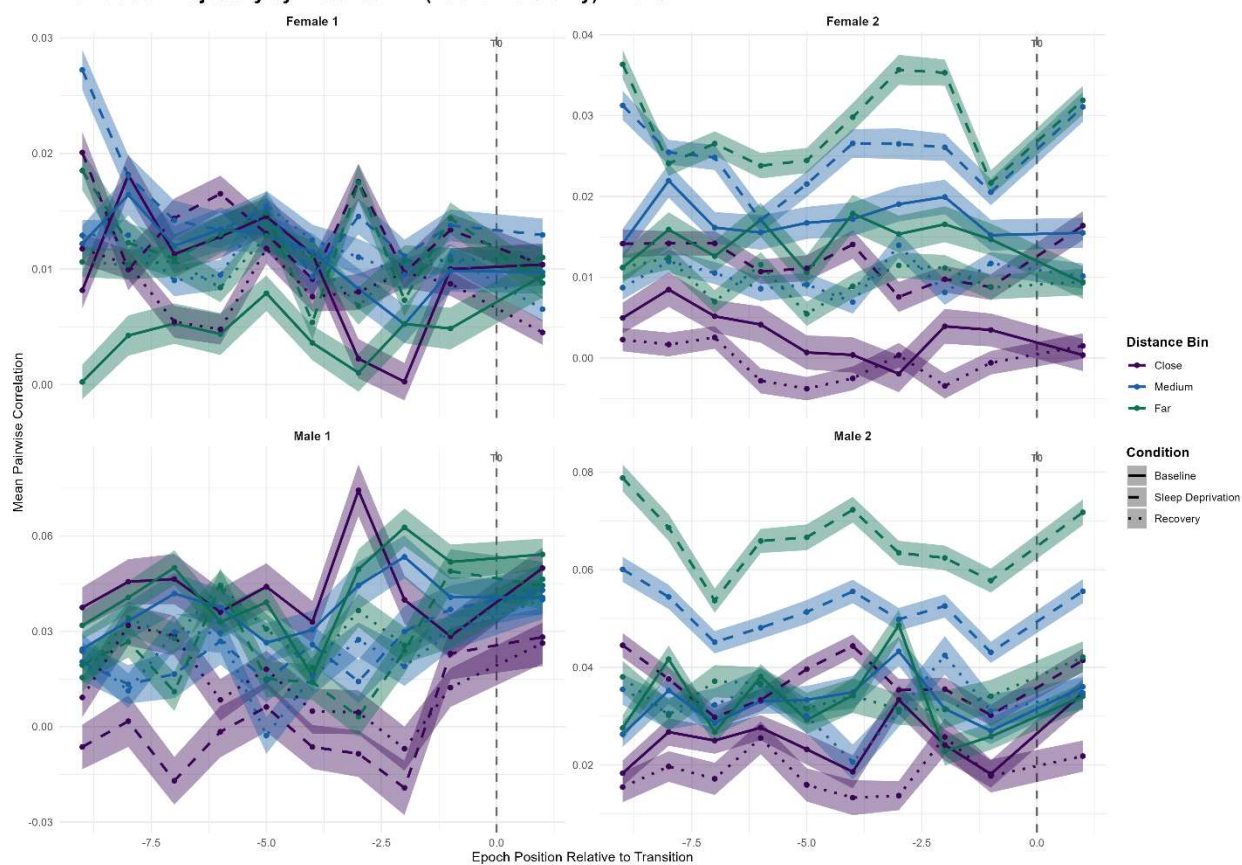
Mean $\pm$ SE pairwise correlation (9 epoch window)



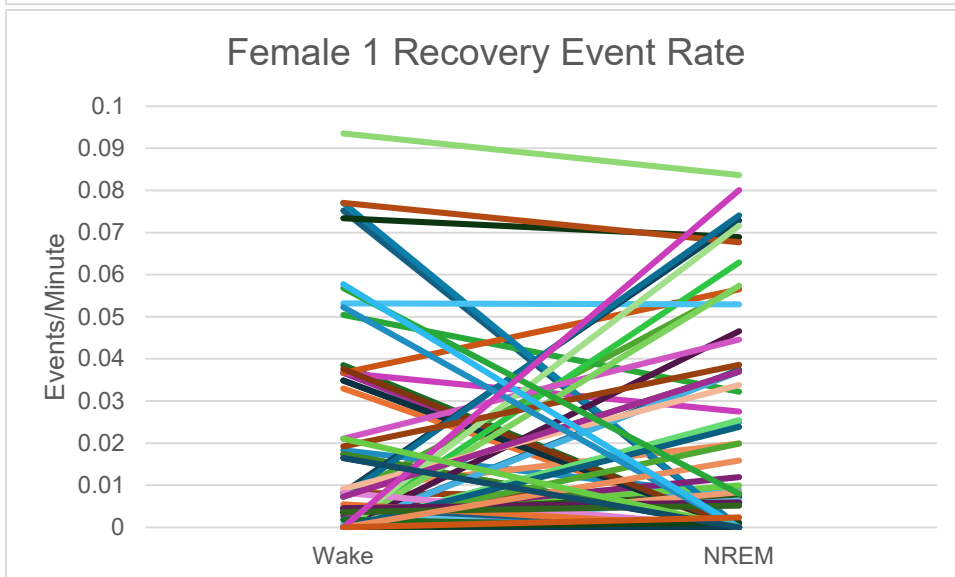
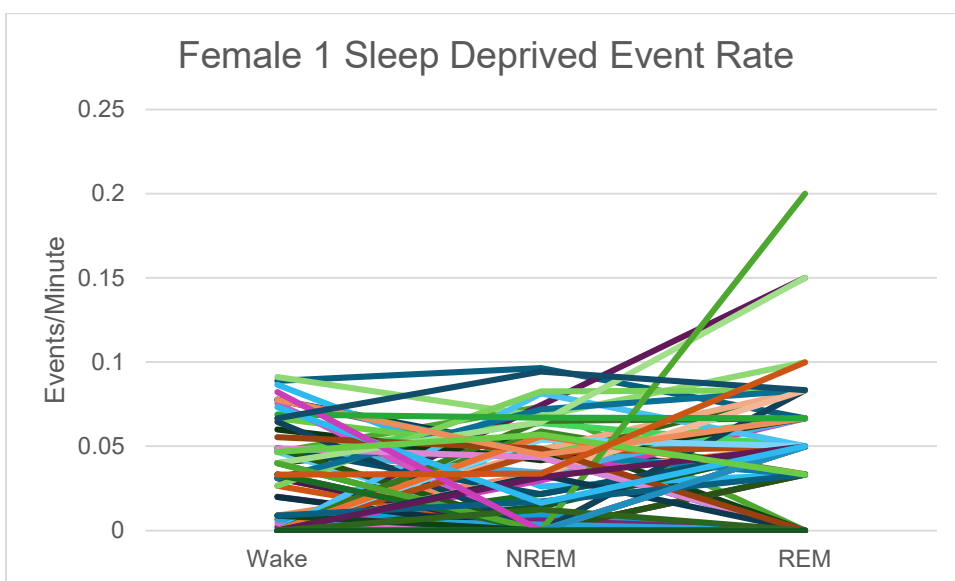
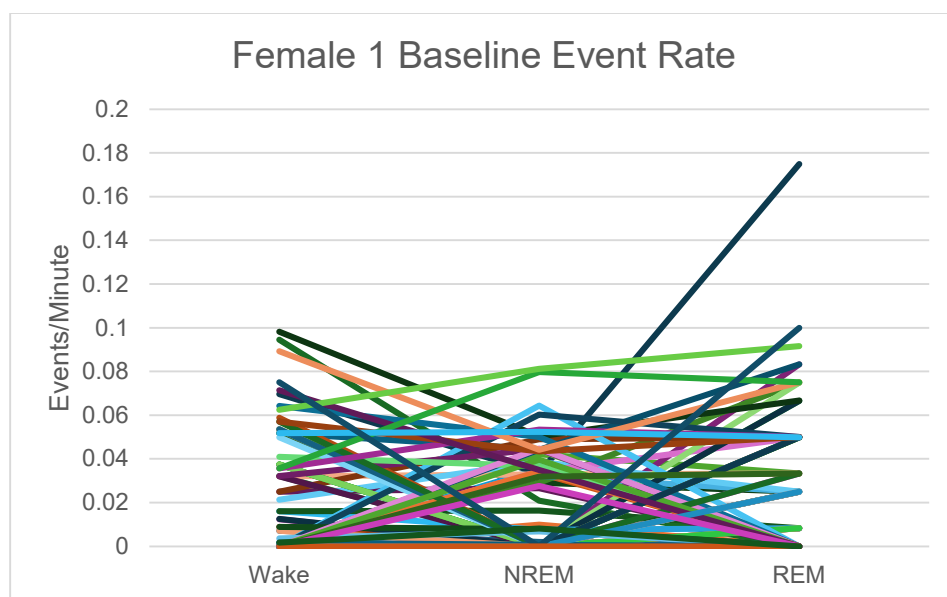
Correlation Trajectory by Distance Bin (Active ROIs Only) - NREM2Wake



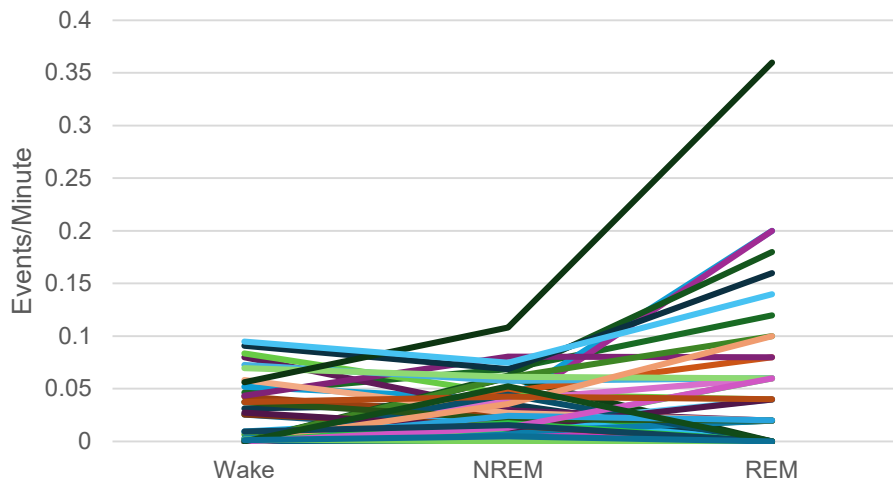
Correlation Trajectory by Distance Bin (Active ROIs Only) - Wake2NREM



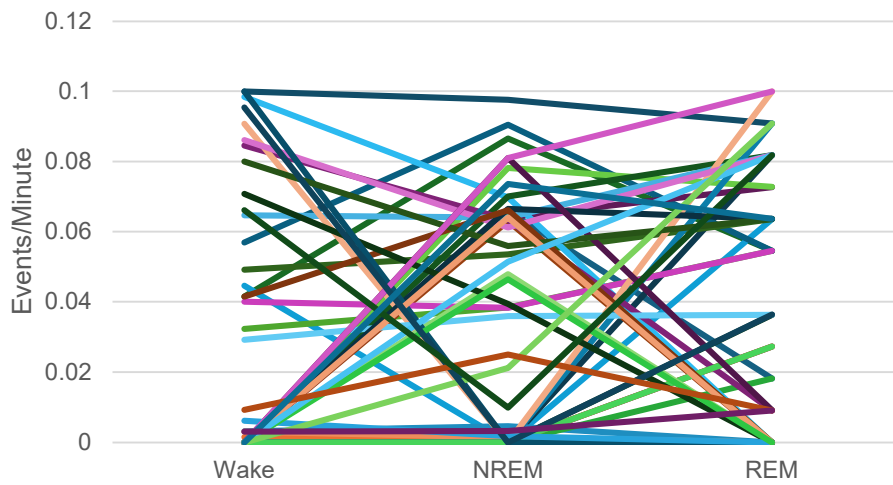
Appendix 8: ROI Event Rates by Sleep Stage



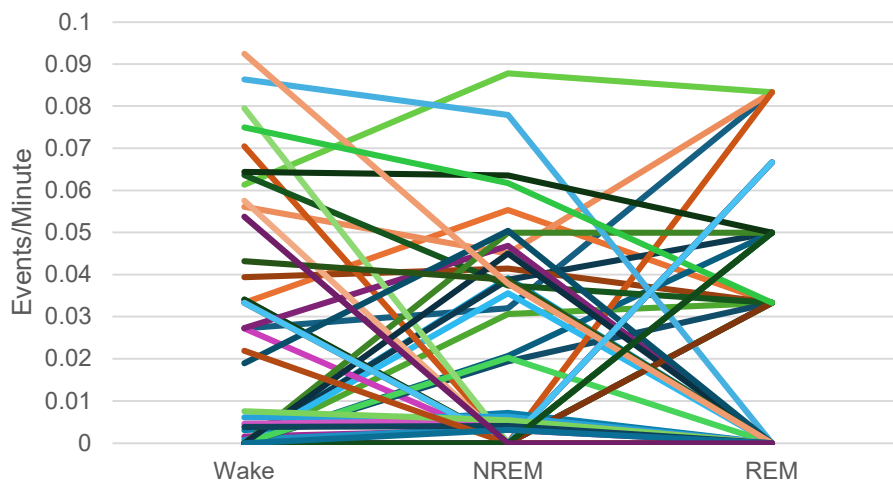
Female 2 Baseline Event Rate

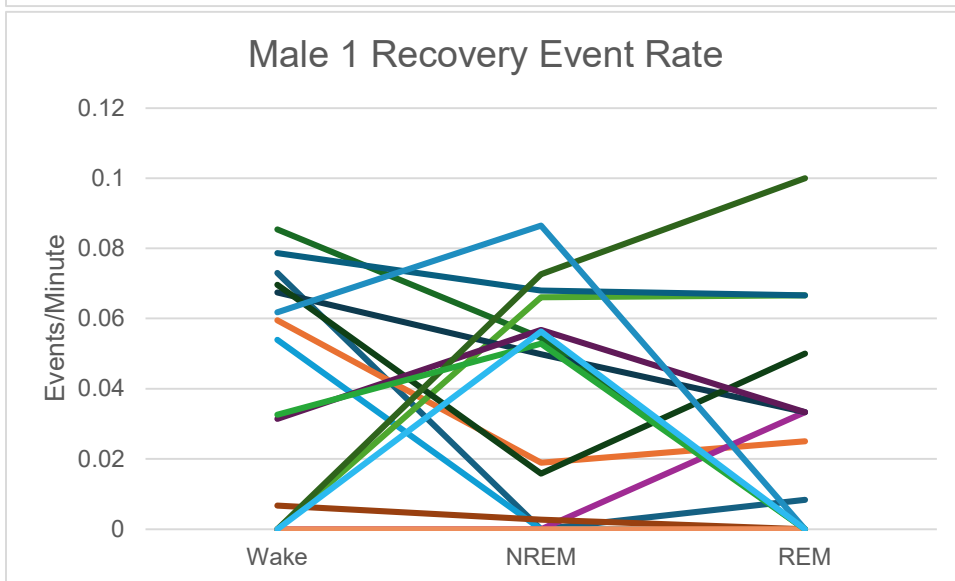
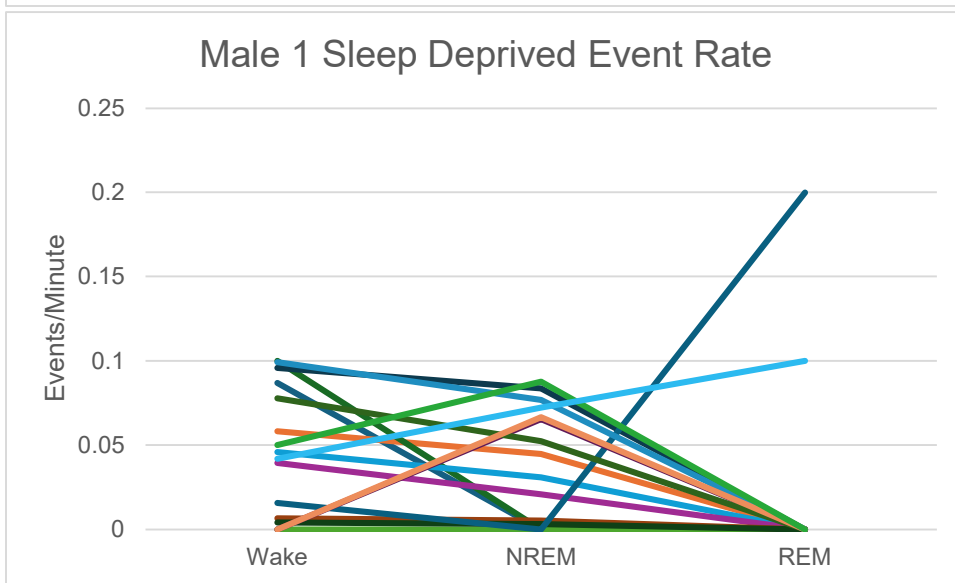
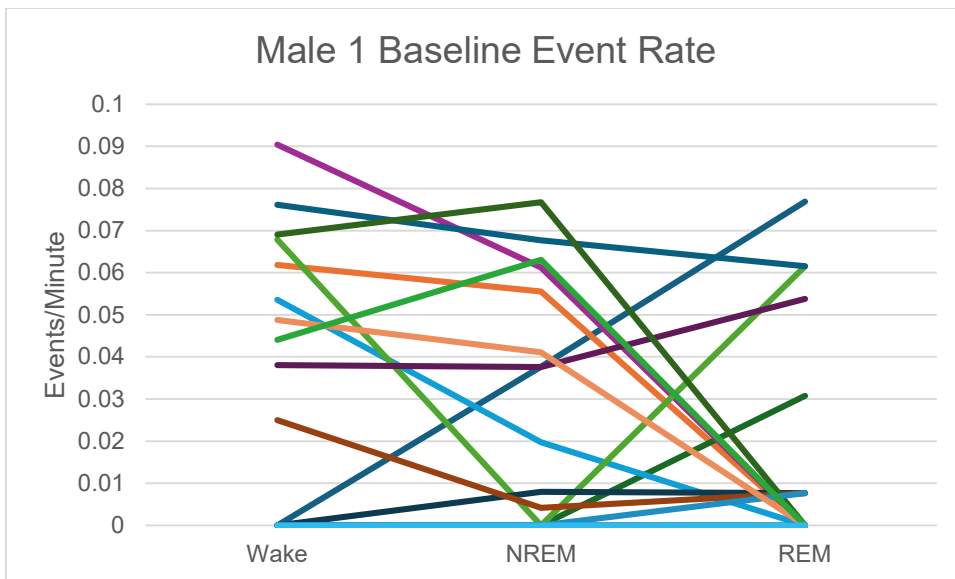


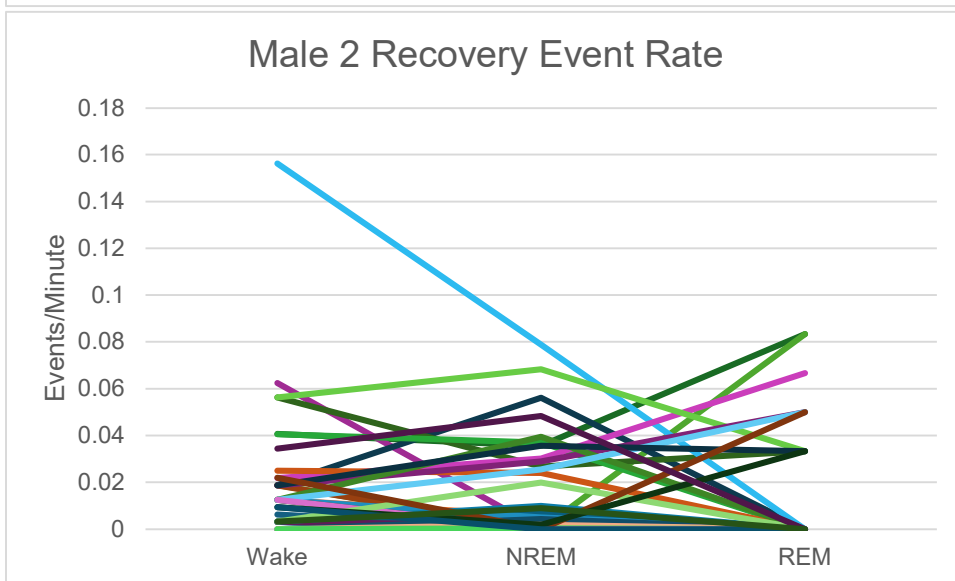
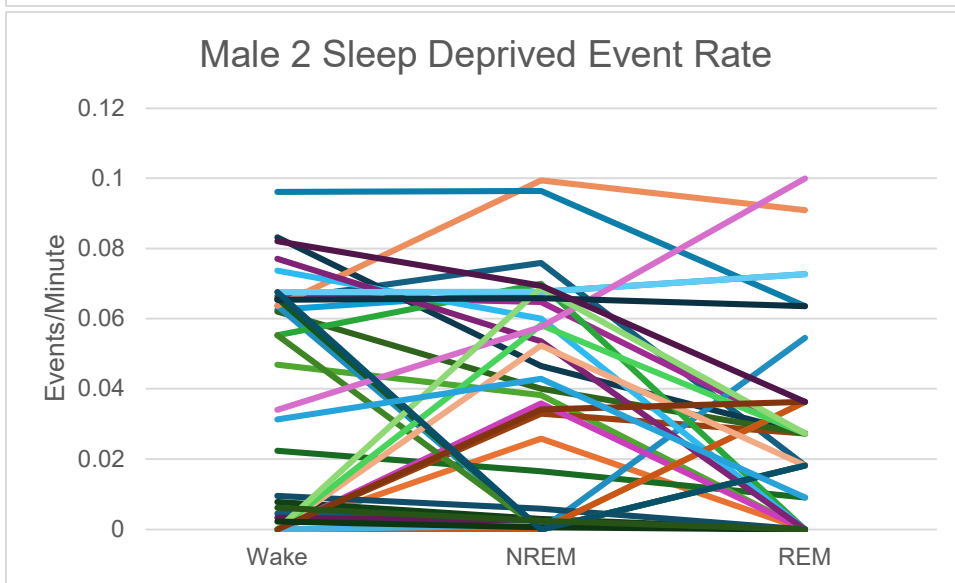
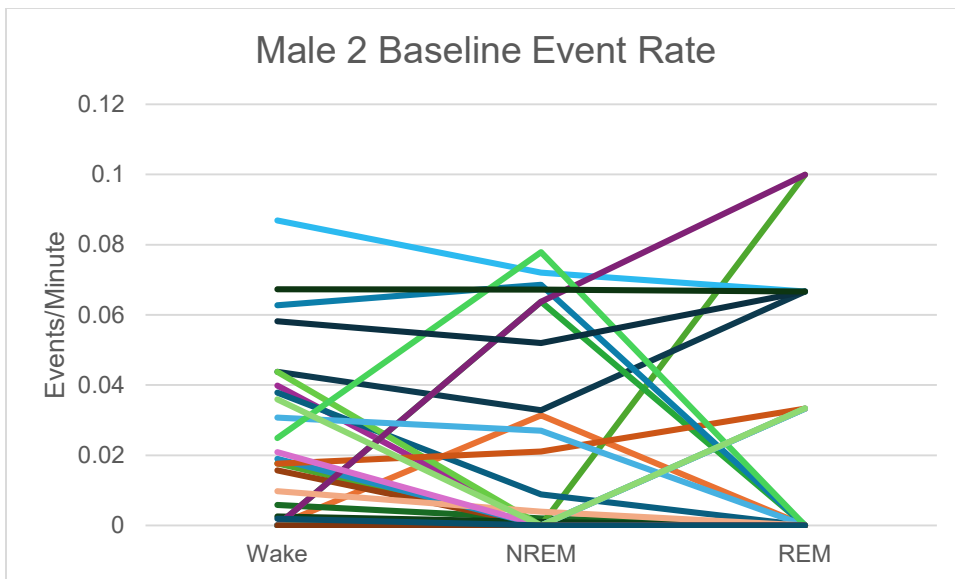
Female 2 Sleep Deprived Event Rate



Female 2 Recovery Event Rate







Appendix 9: Coefficients of Variation

Measure	Within Animal CV	Between Animal CV	Condition
Event Rate	14.1	4.9	Baseline
Correlation (all)	27.6	67.2	Baseline
Close Pairs	54.3	82.3	Baseline
Medium Pairs	26.4	58.7	Baseline
Far Pairs	70.5	91.5	Baseline
Event Rate	13.3	12.7	Sleep Deprived
Correlation (all)	22.8	66	Sleep Deprived
Close Pairs	29.2	69.8	Sleep Deprived
Medium Pairs	22.9	63.2	Sleep Deprived
Far Pairs	29.1	80.3	Sleep Deprived
Event Rate	14	21.9	Recovery
Correlation (all)	23.6	72	Recovery
Close Pairs	96.6	108.1	Recovery
Medium Pairs	23.2	70.1	Recovery
Far Pairs	20	75.3	Recovery